

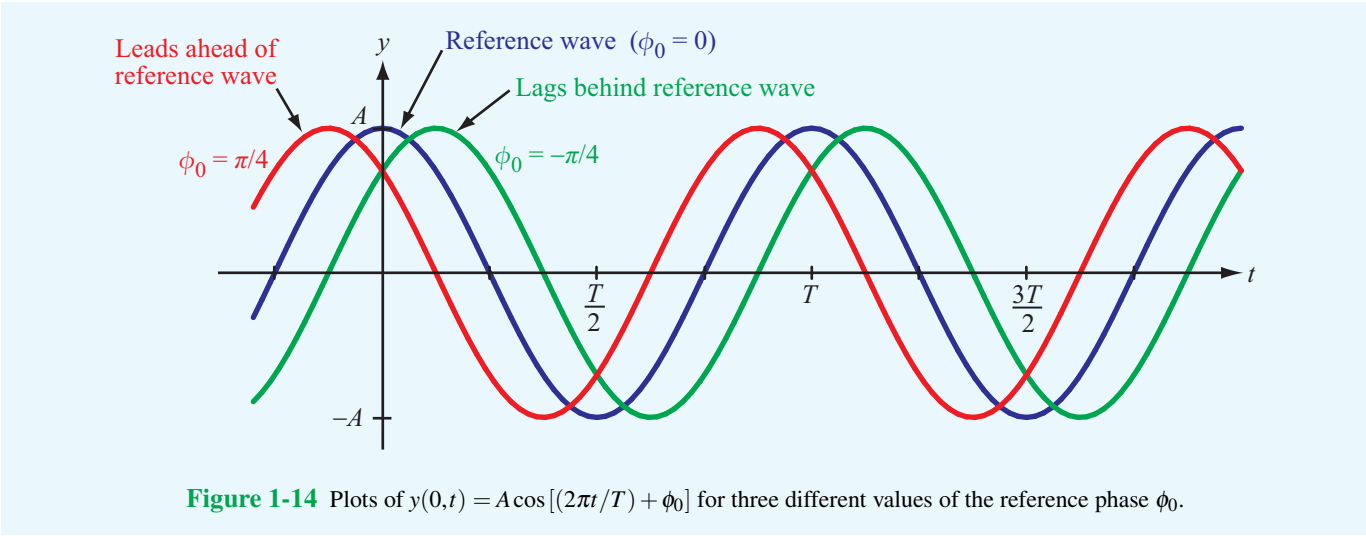


Figure 1-14 Plots of $y(0,t) = A \cos[(2\pi t/T) + \phi_0]$ for three different values of the reference phase ϕ_0 .

proportional to ϕ_0 . This is illustrated by the plots shown in **Fig. 1-14**. We observe that when ϕ_0 is positive, $y(t)$ reaches its peak value, or any other specified value, sooner than when $\phi_0 = 0$. Thus, the wave with $\phi_0 = \pi/4$ is said to **lead** the wave with $\phi_0 = 0$ by a **phase lead** of $\pi/4$; and similarly, the wave with $\phi_0 = -\pi/4$ is said to **lag** the wave with $\phi_0 = 0$ by a **phase lag** of $\pi/4$. A wave function with a negative ϕ_0 takes longer to reach a given value of $y(t)$, such as its peak, than the zero-phase reference function.

► When its value is positive, ϕ_0 signifies a phase lead in time, and when it is negative, it signifies a phase lag. ◀

Exercise 1-3: Consider the red wave shown in **Fig. E1.3**. What is the wave's (a) amplitude, (b) wavelength, and (c) frequency given that its phase velocity is 6 m/s?

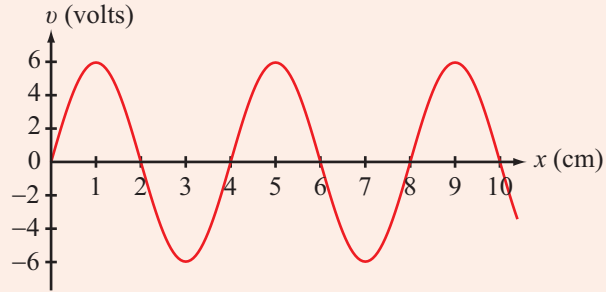


Figure E1.3

Answer: (a) $A = 6$ V, (b) $\lambda = 4$ cm, (c) $f = 150$ Hz. (See **EM**.)

Exercise 1-4: The wave shown in red in **Fig. E1.4** is given by $v = 5 \cos 2\pi t/8$. Of the following four equations:

- (1) $v = 5 \cos(2\pi t/8 - \pi/4)$,
- (2) $v = 5 \cos(2\pi t/8 + \pi/4)$,
- (3) $v = -5 \cos(2\pi t/8 - \pi/4)$,
- (4) $v = 5 \sin 2\pi t/8$,

(a) which equation applies to the green wave? (b) which equation applies to the blue wave?

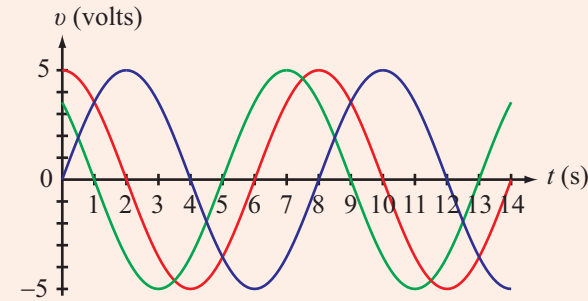


Figure E1.4

Answer: (a) #2, (b) #4. (See **EM**.)

相位超前、滞后与波形练习

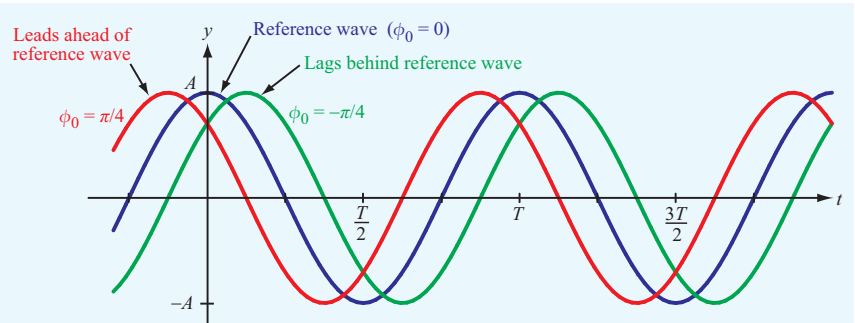


图1-14 $y(0,t) = A \cos(2\pi t/T + \phi_0)$ 在三种参考相位下的时间波形。

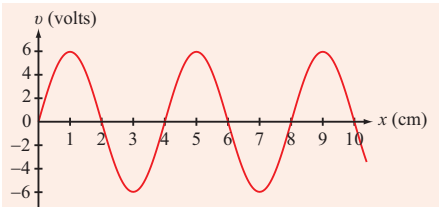
(承上页) 在固定位置, $\phi_0 > 0$ 的波比 $\phi_0 = 0$ 的参考波更早达到同一相位状态。例如 $\phi_0 = \pi/4$ 表示相位超前 $\pi/4$; $\phi_0 = -\pi/4$ 表示相位滞后 $\pi/4$ 。相对于零相位参考波, 负参考相位需要更晚的时刻才能达到相应波峰。

图中Leads ahead: 超前; Lags behind: 滞后; Reference wave: 参考波。此处“超前 / 滞后”针对固定位置、采用 $\cos(\omega t + \phi_0)$ 及 $\omega > 0$ 的时间约定; 不能脱离约定只看相位常数正负。

练习1-3

图E1.3给出红色波形, 已知相速度为6 m/s, 求 (a) 振幅; (b) 波长; (c) 频率。

答案: $A = 6$ V, $\lambda = 4$ cm = 0.04 m, $f = u_p/\lambda = 6/0.04 = 150$ Hz。

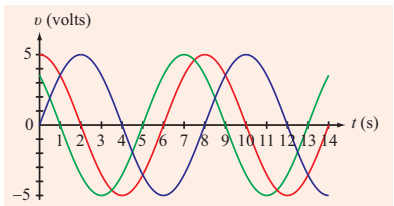


图E1.3 横轴x (cm), 纵轴电压v (V)。

练习1-4

图E1.4中的红色波形为 $v = 5 \cos(2\pi t/8)$ 。以下哪个表达式分别对应 (a) 绿色波形; (b) 蓝色波形?

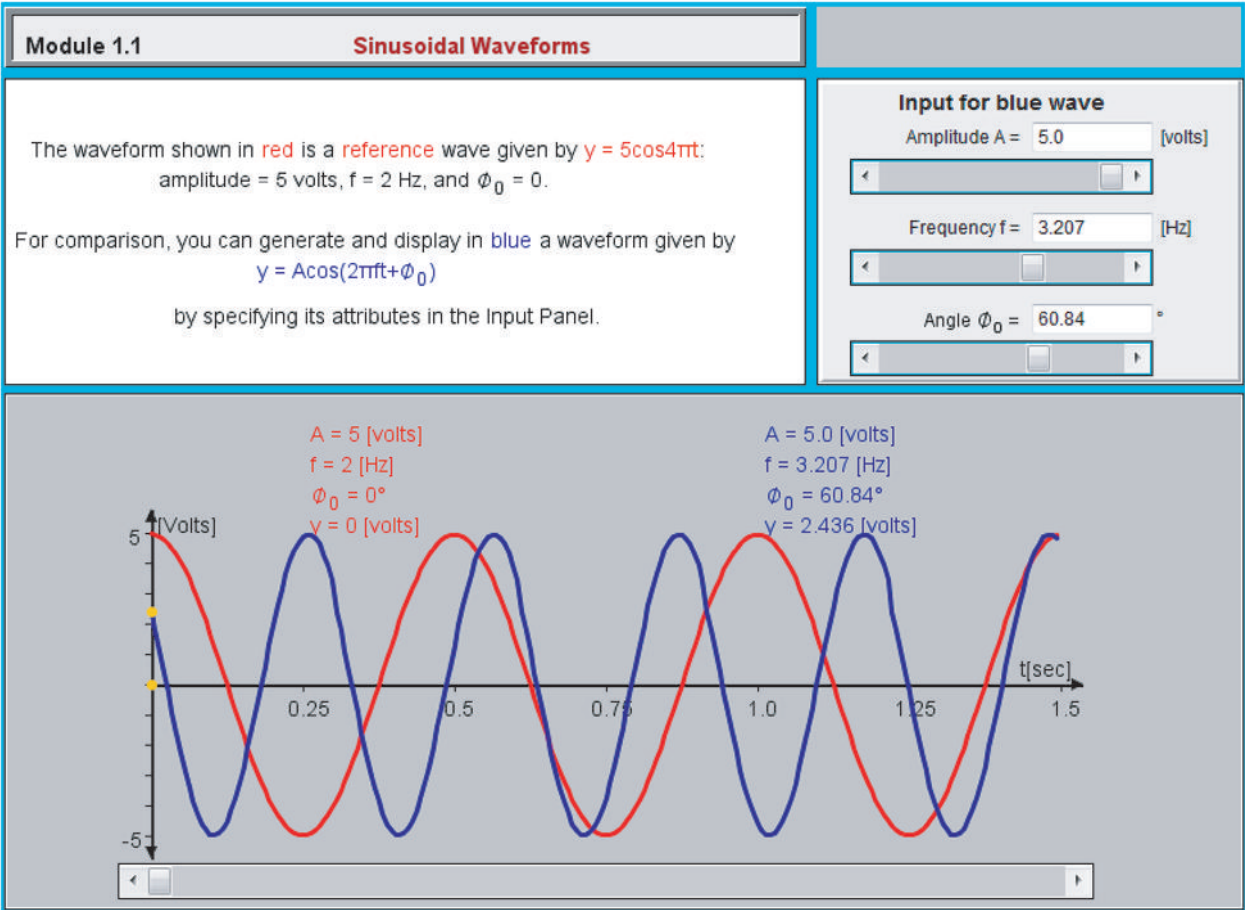
- (1) $v = 5 \cos(2\pi t/8 - \pi/4)$,
- (2) $v = 5 \cos(2\pi t/8 + \pi/4)$,
- (3) $v = -5 \cos(2\pi t/8 - \pi/4)$,
- (4) $v = 5 \sin 2\pi t/8$,



图E1.4 横轴t (s), 纵轴电压v (V)。

答案: (a) 第 (2) 式, 超前参考余弦波 $\pi/4$; (b) 第 (4) 式, $5 \sin(2\pi t/8)$ 。详解参见原书配套EM资源。

Module 1.1 Sinusoidal Waveforms Learn how the shape of the waveform is related to the amplitude, frequency, and reference phase angle of a sinusoidal wave.



Exercise 1-5: The electric field of a traveling electromagnetic wave is given by

$$E(z, t) = 10 \cos(\pi \times 10^7 t + \pi z / 15 + \pi / 6) \quad (\text{V/m}).$$

Determine (a) the direction of wave propagation, (b) the wave frequency f , (c) its wavelength λ , and (d) its phase velocity u_p .

Answer: (a) $-z$ direction, (b) $f = 5$ MHz, (c) $\lambda = 30$ m, (d) $u_p = 1.5 \times 10^8$ m/s. (See [EM](#).)

1-4.2 Sinusoidal Waves in a Lossy Medium

If a wave is traveling in the x direction in a **lossy medium**, its amplitude decreases as $e^{-\alpha x}$. This factor is called the **attenuation factor**, and α is called the **attenuation constant** of the medium and its unit is neper per meter (Np/m). Thus, in general,

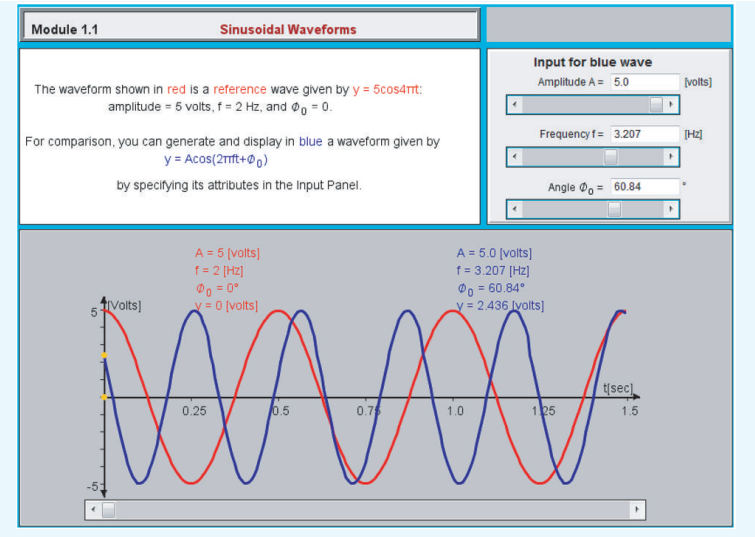
$$y(x, t) = Ae^{-\alpha x} \cos(\omega t - \beta x + \phi_0). \quad (1.33)$$

The wave amplitude is now $Ae^{-\alpha x}$, not just A . **Figure 1-15** shows a plot of $y(x, t)$ as a function of x at $t = 0$ for $A = 10$ m, $\lambda = 2$ m, $\alpha = 0.2$ Np/m, and $\phi_0 = 0$. Note that the envelope of the wave pattern decreases as $e^{-\alpha x}$.

正弦波练习与有损介质

模块1.1 正弦波形

通过配套交互模块，观察正弦波的振幅、频率和参考相位角如何影响波形。下图保留原书模块界面。



原书模块1.1：参数调节与正弦波形显示。

练习1-5

某行进电磁波的电场为：

$$E(z, t) = 10 \cos(\pi \times 10^7 t + \pi z / 15 + \pi / 6) \quad (\text{V/m}).$$

求 (a) 传播方向； (b) 频率 f ； (c) 波长 λ ； (d) 相速度 u_p 。
答案： t 、 z 两项同号，波沿 $-z$ 方向传播； $\omega = \pi \times 10^7$ rad/s，故 $f = \omega / (2\pi) = 5$ MHz； $\beta = \pi / 15$ rad/m，故 $\lambda = 2\pi / \beta = 30$ m； $u_p = f\lambda = 1.5 \times 10^8$ m/s。

1-4.2 有损介质中的正弦波

波在有损介质中沿 $+x$ 方向传播时，振幅随传播距离按因子 $e^{-\alpha x}$ 衰减。 $e^{-\alpha x}$ 称为衰减因子， α 为衰减常数，单位是奈培每米 (Np/m)。因此一般可写为：

$$y(x, t) = Ae^{-\alpha x} \cos(\omega t - \beta x + \phi_0). \quad (1.33)$$

此时振幅为 $Ae^{-\alpha x}$ ，而非 A ，更不能把振幅直接写成无量纲的 $e^{-\alpha x}$ 。图1-15给出 $A = 10$ m、 $\lambda = 2$ m、 $\alpha = 0.2$ Np/m、 $\phi_0 = 0$ 时，在 $t = 0$ 的波形；其正、负包络分别为 $\pm 10e^{-0.2x}$ 。

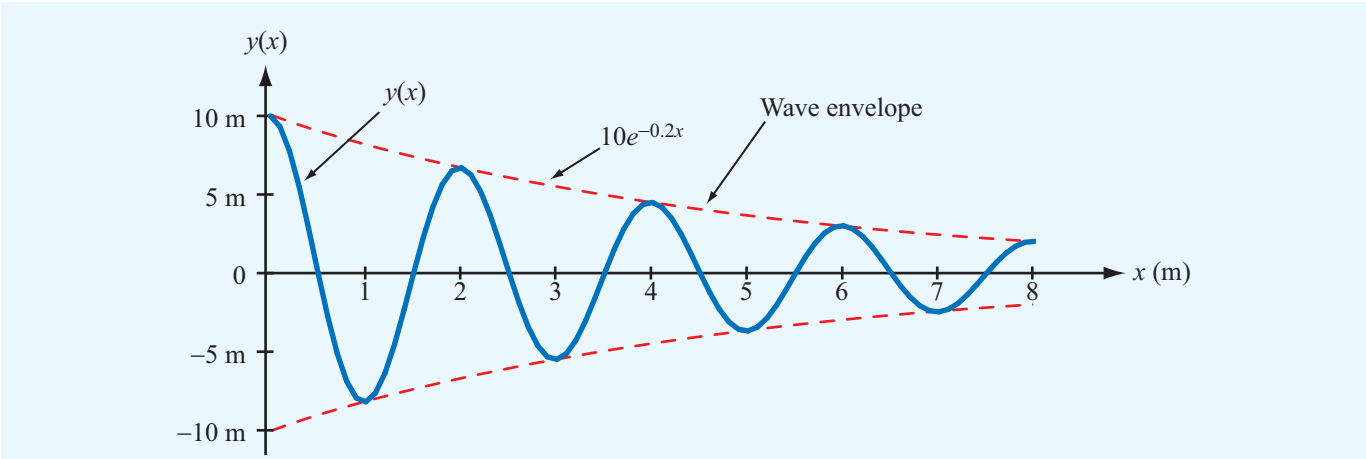


Figure 1-15 Plot of $y(x) = (10e^{-0.2x} \cos \pi x)$ meters. Note that the envelope is bounded between the curve given by $10e^{-0.2x}$ and its mirror image.

The real unit of α is (1/m); the neper (Np) part is a dimensionless, artificial adjective traditionally used as a reminder that the unit (Np/m) refers to the attenuation constant of the medium, α . A similar practice is applied to the phase constant β by assigning it the unit (rad/m) instead of just (1/m).

Concept Question 1-6: How can you tell if a wave is traveling in the positive x direction or the negative x direction?

Concept Question 1-7: How does the envelope of the wave pattern vary with distance in (a) a lossless medium and (b) a lossy medium?

Concept Question 1-8: Why does a negative value of ϕ_0 signify a phase lag?

Example 1-1: Sound Wave in Water

An acoustic wave traveling in the x direction in a fluid (liquid or gas) is characterized by a differential pressure $p(x, t)$. The unit for pressure is newton per square meter (N/m²). Find an expression for $p(x, t)$ for a sinusoidal sound wave traveling in the positive x direction in water given that the wave frequency

is 1 kHz, the velocity of sound in water is 1.5 km/s, the wave amplitude is 10 N/m², and $p(x, t)$ was observed to be at its maximum value at $t = 0$ and $x = 0.25$ m. Treat water as a lossless medium.

Solution: According to the general form given by Eq. (1.17) for a wave traveling in the positive x direction,

$$p(x, t) = A \cos \left(\frac{2\pi}{T} t - \frac{2\pi}{\lambda} x + \phi_0 \right) \quad (\text{N/m}^2).$$

The amplitude $A = 10$ N/m², $T = 1/f = 10^{-3}$ s, and from $u_p = f\lambda$,

$$\lambda = \frac{u_p}{f} = \frac{1.5 \times 10^3}{10^3} = 1.5 \text{ m}.$$

Hence,

$$p(x, t) = 10 \cos \left(2\pi \times 10^3 t - \frac{4\pi}{3} x + \phi_0 \right) \quad (\text{N/m}^2).$$

Since at $t = 0$ and $x = 0.25$ m, $p(0.25, 0) = 10$ N/m², we have

$$10 = 10 \cos \left(\frac{-4\pi}{3} 0.25 + \phi_0 \right) = 10 \cos \left(\frac{-\pi}{3} + \phi_0 \right),$$

which yields the result $(\phi_0 - \pi/3) = \cos^{-1}(1)$, or $\phi_0 = \pi/3$. Hence,

$$p(x, t) = 10 \cos \left(2\pi \times 10^3 t - \frac{4\pi}{3} x + \frac{\pi}{3} \right) \quad (\text{N/m}^2).$$

衰减包络与水中声波示例

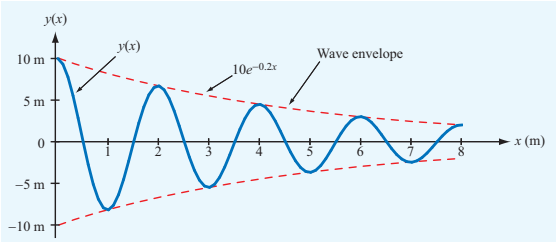


图1-15 $y(x) = 10e^{-0.2x} \cos(\pi x)$ m; 上下包络为 $\pm 10e^{-0.2x}$ m。

从量纲看， α 的单位为 m^{-1} 。奈培（Np）用于标明其对应以自然对数定义的衰减量；Np/m强调这是衰减常数。类似地， β 采用rad/m，以区分单位长度的相位变化与一般的倒长度量。

概念题1-6：如何判断波沿+x还是−x方向传播？

概念题1-7：排除几何扩散后，波的包络在（a）无损介质；（b）有损介质中如何随距离变化？

概念题1-8：为什么在本书时间约定下，负 ϕ_0 表示相位滞后？

例1-1 水中的声波

流体中的声波可用相对静压的声压扰动 $p(x,t)$ 描述，单位为帕斯卡，即N/m²。已知水中正弦声波沿+x方向传播，频率1 kHz，声速1.5 km/s，声压振幅10 N/m²，且在 $t = 0$ 、 $x = 0.25$ m处达到正最大值。将水视为无损介质，求 $p(x,t)$ 。

解：按式（1.17）的形式：

$$p(x, t) = A \cos \left(\frac{2\pi}{T} t - \frac{2\pi}{\lambda} x + \phi_0 \right) \quad (\text{N/m}^2).$$

$A = 10$ N/m²， $T = 1/f = 10^{-3}$ s。由 $u_p = f\lambda$ 得：

$$\lambda = \frac{u_p}{f} = \frac{1.5 \times 10^3}{10^3} = 1.5 \text{ m}.$$

代入周期和波长：

$$p(x, t) = 10 \cos \left(2\pi \times 10^3 t - \frac{4\pi}{3} x + \phi_0 \right) \quad (\text{N/m}^2).$$

将 $p(0.25, 0) = 10$ N/m²代入，有：

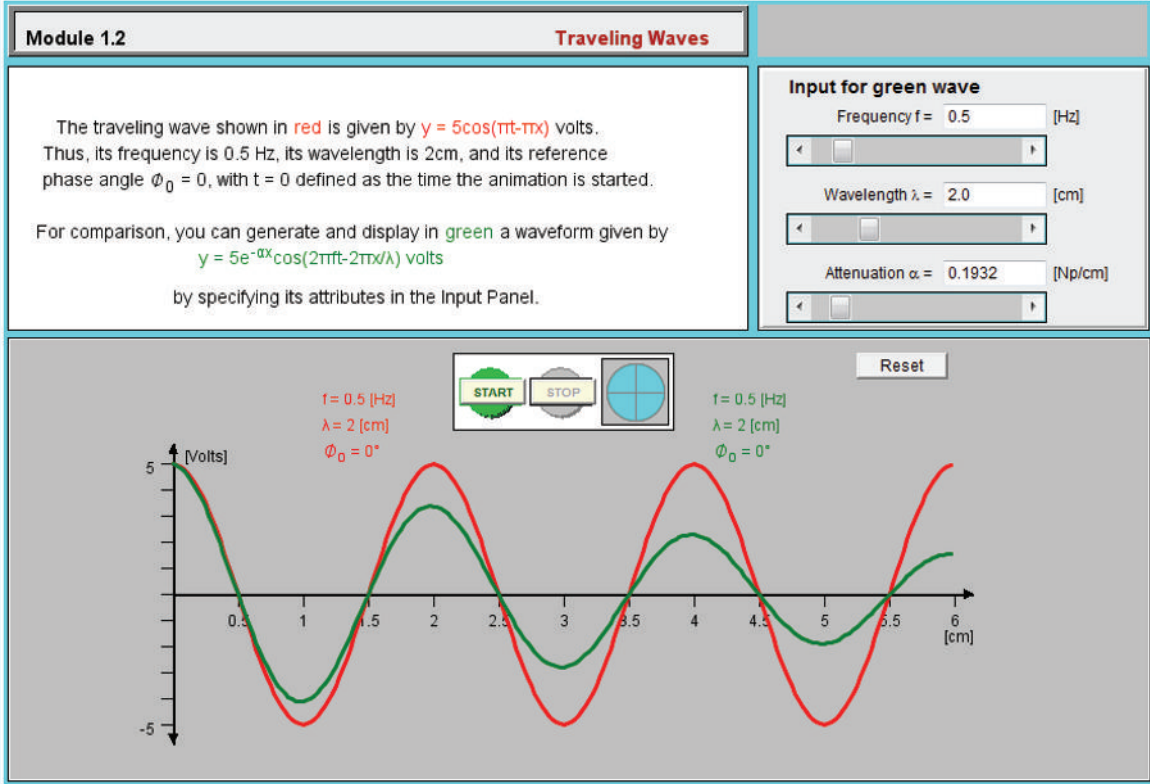
$$10 = 10 \cos \left(\frac{-4\pi}{3} 0.25 + \phi_0 \right) = 10 \cos \left(\frac{-\pi}{3} + \phi_0 \right),$$

因此 $\phi_0 - \pi/3 = 2n\pi$ ；取代表值 $\phi_0 = \pi/3$ ，最终得到：

$$p(x, t) = 10 \cos \left(2\pi \times 10^3 t - \frac{4\pi}{3} x + \frac{\pi}{3} \right) \quad (\text{N/m}^2).$$

核算： $\lambda = 1500/1000 = 1.5$ m， $\beta = 2\pi/\lambda = 4\pi/3$ rad/m； $x = 0.25$ m、 $t = 0$ 时相位为零，声压确为+10 N/m²。原书给出的结果通过代入检查。

Module 1.2 Traveling Waves Learn how the shape of a traveling wave is related to its frequency and wavelength and to the attenuation constant of the medium.



Example 1-2: Power Loss

A laser beam of light propagating through the atmosphere is characterized by an electric field given by

$$E(x,t) = 150e^{-0.03x} \cos(3 \times 10^{15}t - 10^7x) \quad (\text{V/m}),$$

where x is the distance from the source in meters. The attenuation is due to absorption by atmospheric gases. Determine

- (a) the direction of wave travel,
- (b) the wave velocity, and
- (c) the wave amplitude at a distance of 200 m.

Solution: (a) Since the coefficients of t and x in the argument of the cosine function have opposite signs, the wave must be traveling in the $+x$ direction.

(b)

$$u_p = \frac{\omega}{\beta} = \frac{3 \times 10^{15}}{10^7} = 3 \times 10^8 \text{ m/s},$$

and this is equal to c , which is the velocity of light in free space.

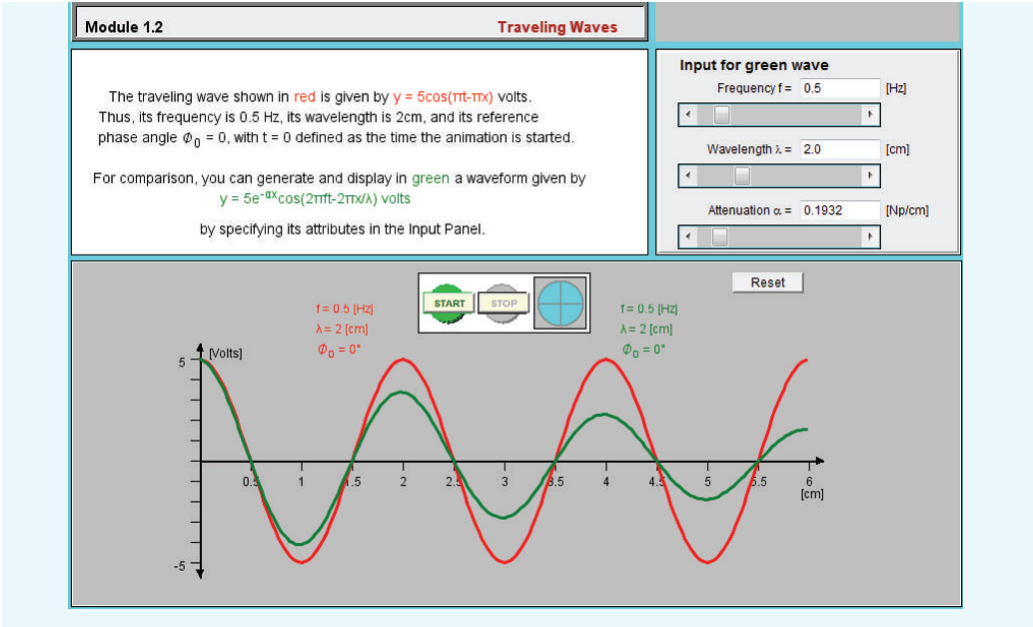
(c) At $x = 200$ m, the amplitude of $E(x,t)$ is

$$150e^{-0.03 \times 200} = 0.37 \quad (\text{V/m}).$$

Exercise 1-6: Consider the red wave shown in **Fig. E1.6**. What is the wave's (a) amplitude (at $x = 0$), (b) wavelength, and (c) attenuation constant?

模块1.2与例1-2：功率损耗

模块1.2 行波：观察行波的频率、波长以及介质衰减常数对波形的影响。



原书模块1.2：行波传播、相位与衰减。

例1-2 功率损耗

一束激光在大气中传播，其电场为：

$$E(x,t) = 150e^{-0.03x} \cos(3 \times 10^{15}t - 10^7x) \quad (\text{V/m}),$$

x 是距光源的距离，单位为米。大气气体吸收导致衰减。求 (a) 波的传播方向； (b) 波速； (c) 距光源200 m处的电场振幅。

解： (a) 余弦相位中 t 项与 x 项异号，因此沿 $+x$ 方向传播。

$$u_p = \frac{\omega}{\beta} = \frac{3 \times 10^{15}}{10^7} = 3 \times 10^8 \text{ m/s},$$

(b) 上述相速度约等于真空光速 c 。

(c) $x = 200$ m处的电场振幅为：

$$150e^{-0.03 \times 200} = 0.37 \quad (\text{V/m}).$$

数值复核： $150e^{-6} = 0.3718 \text{ V/m}$ ，按原书精度为0.37 V/m。Power loss应译为“功率损耗”，不能译为“断电”；本例具体所求为场的振幅。

练习1-6

观察下页图E1.6的红色波形，求 (a) $x = 0$ 处振幅； (b) 波长； (c) 衰减常数。

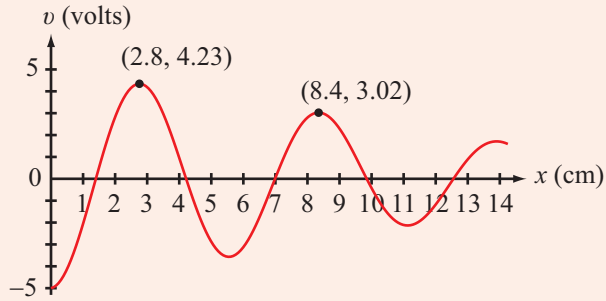


Figure E1.6

Answer: (a) 5 V, (b) 5.6 cm, (c) $\alpha = 0.06$ Np/cm. (See [EM](#).)

Exercise 1-7: The red wave shown in [Fig. E1.7](#) is given by $v = 5 \cos 4\pi x$ (V). What expression is applicable to (a) the blue wave and (b) the green wave?

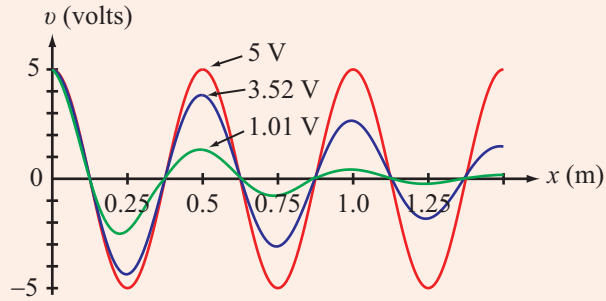


Figure E1.7

Answer: (a) $v = 5e^{-0.7x} \cos 4\pi x$ (V), (b) $v = 5e^{-3.2x} \cos 4\pi x$ (V). (See [EM](#).)

Exercise 1-8: An electromagnetic wave is propagating in the z direction in a lossy medium with attenuation constant $\alpha = 0.5$ Np/m. If the wave's electric-field amplitude is 100 V/m at $z = 0$, how far can the wave travel before its amplitude is reduced to (a) 10 V/m, (b) 1 V/m, and (c) 1 μ V/m?

Answer: (a) 4.6 m, (b) 9.2 m, (c) 37 m. (See [EM](#).)

electromagnetic spectrum ([Fig. 1-16](#)). Other members of this family include gamma rays, X-rays, infrared waves, and radio waves. Generically, they all are called EM waves because they share the following fundamental properties:

- A **monochromatic** (single frequency) EM wave consists of electric and magnetic fields that oscillate at the same frequency f .
- The phase velocity of an EM wave propagating in a vacuum is a universal constant given by the velocity of light c , defined earlier by Eq. (1.14).
- In a vacuum, the wavelength λ of an EM wave is related to its oscillation frequency f by

$$\lambda = \frac{c}{f}. \quad (1.34)$$

Whereas all monochromatic EM waves share these properties, each is distinguished by its own wavelength λ , or equivalently by its own oscillation frequency f .

The visible part of the EM spectrum shown in [Fig. 1-16](#) covers a very narrow wavelength range extending between $\lambda = 0.4 \mu\text{m}$ (violet) and $\lambda = 0.7 \mu\text{m}$ (red). As we move progressively toward shorter wavelengths, we encounter the ultraviolet, X-ray, and gamma-ray bands, each so named because of historical reasons associated with the discovery of waves with those wavelengths. On the other side of the visible spectrum lie the infrared band and then the microwave part of the radio region. Because of the link between λ and f given by Eq. (1.34), each of these spectral ranges may be specified in terms of its wavelength range or its frequency range. In practice, however, a wave is specified in terms of its wavelength λ if $\lambda < 1$ mm, which encompasses all parts of the EM spectrum except for the radio region, and the wave is specified in terms of its frequency f if $\lambda > 1$ mm (i.e., in the radio region). A wavelength of 1 mm corresponds to a frequency of 3×10^{11} Hz = 300 GHz in free space.

The radio spectrum consists of several individual bands, as shown in the chart of [Fig. 1-17](#) (see p. 32). Each band covers one decade of the radio spectrum and has a letter designation based on a nomenclature defined by the International Telecommunication Union. Waves of different frequencies have different applications because they are excited by different mechanisms, and the properties of an EM wave propagating in a nonvacuum material may vary considerably from one band to another.

Although no precise definition exists for the extent of the **microwave band**, it is conventionally regarded to cover the full ranges of the UHF, SHF, and EHF bands. The EHF band is sometimes referred to as the **millimeter-wave band** because

衰减练习与1-5电磁波谱

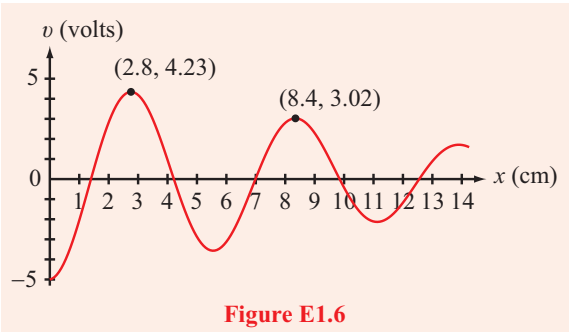


Figure E1.6

图E1.6, 横轴x (cm), 纵轴电压v (V)。

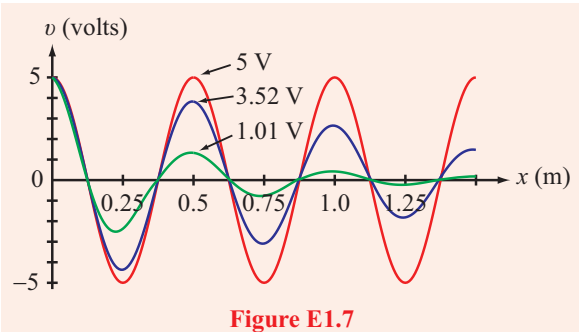


Figure E1.7

图E1.7, 横轴x (m), 纵轴电压v (V)。

练习1-6答案: (a) 5 V; (b) $\lambda = 5.6$ cm; (c) 由两个包络采样值, $\alpha = \ln(4.23/3.02)/(8.4-2.8) \approx 0.060$ Np/cm。

练习1-7: 图E1.7红色波形为 $v = 5 \cos(4\pi x)$ V, 求蓝色、绿色波形的表达式。

答案: 蓝色: $v = 5e^{-0.7x} \cos(4\pi x)$ V; 绿色: $v = 5e^{-3.2x} \cos(4\pi x)$ V, 其中x以米计。

练习1-8: 电磁波沿+z方向在有损介质中传播, $\alpha = 0.5$ Np/m, $z = 0$ 处电场振幅为100 V/m。振幅分别降到10 V/m、1 V/m和1 μ V/m时, 传播距离是多少?

答案: 由 $z = \ln(100/A)/0.5$ 得到约4.6 m、9.2 m和37 m。第三问1 μ V/m = 10^{-6} V/m。

1-5 电磁波谱

按频率或波长连续排列的电磁波构成电磁波谱 (图1-16), 包括伽马射线、X射线、紫外线、可见光、红外线和无线电波。它们具有以下共同性质: 单色波的电场与磁场以同一频率振荡; 在真空中其相速度为式 (1.14) 的 c ; 真空波长满足:

$$\lambda = \frac{c}{f}. \quad (1.34)$$

虽然具有共同性质, 各单色波仍可由各自的频率或真空波长区分。原书给出可见光范围约0.4 μm (紫) 至0.7 μm (红)。更短波长依次属于紫外、X射线和伽马射线; 更长波长则进入红外与无线电波段。这些名称与发现历史有关。

任何波段均可用频率或波长表示。原书采用的惯例是: 波长小于1 mm时常用波长, 大于1 mm时常用频率。真空中1 mm对应300 GHz。

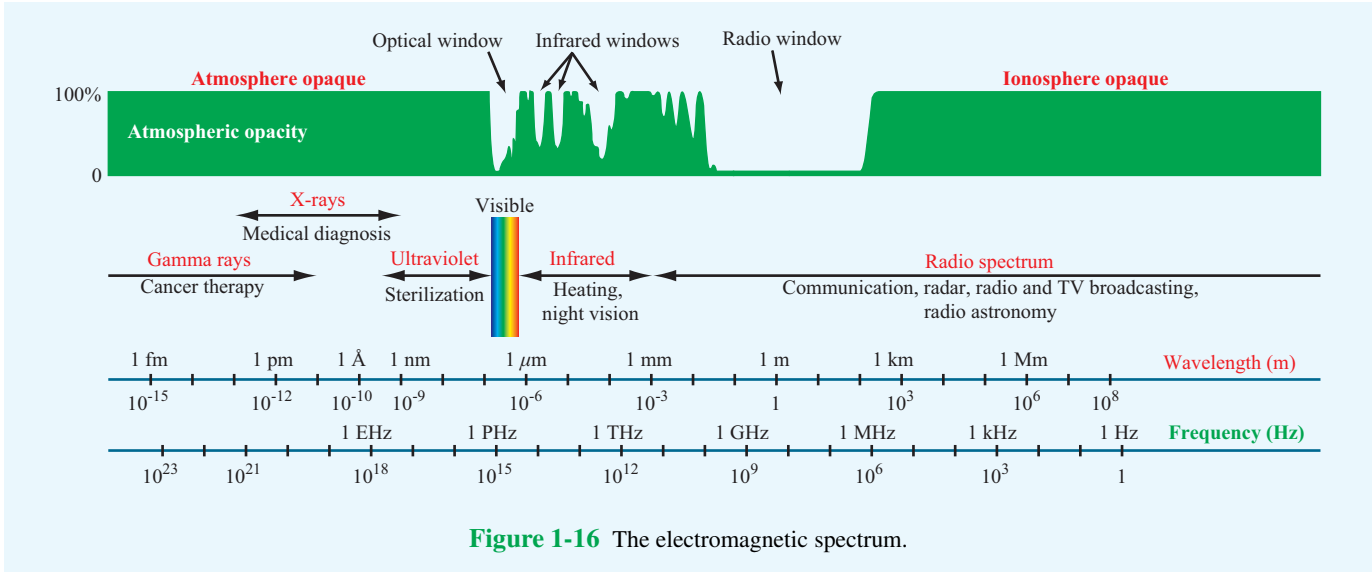
图1-17将无线电波谱分为若干频段。各标准频段覆盖一个**十倍频程** (上下限频率相差10倍), 其字母名称采用国际电信联盟的命名。不同频段的产生机制、介质传播性质及应用有所不同。

微波频段没有唯一边界定义; 原书采用UHF、SHF、EHF三个频段合计的范围。EHF也称毫米波段, 波长为1 mm至1 cm, 对应300 GHz至30 GHz。

核算: 练习1-8三个距离的未舍入值分别约4.605、9.210、36.841 m。原书频段划分图属于历史资料。

1-5 The Electromagnetic Spectrum

Visible light belongs to a family of waves arranged according to frequency and wavelength along a continuum called the



the wavelength range covered by this band extends from 1 mm (300 GHz) to 1 cm (30 GHz).

- Concept Question 1-9:** What are the three fundamental properties of EM waves?
- Concept Question 1-10:** What is the range of frequencies covered by the microwave band?
- Concept Question 1-11:** What is the wavelength range of the visible spectrum? What are some of the applications of the infrared band?

1-6 Review of Complex Numbers

Any **complex number** z can be expressed in **rectangular form** as

$$z = x + jy, \tag{1.35}$$

where x and y are the **real** ($\Re$) and **imaginary** ($\Im$) parts of z , respectively, and $j = \sqrt{-1}$. That is,

$$x = \Re(z), \quad y = \Im(z). \tag{1.36}$$

Alternatively, z may be cast in **polar form** as

$$z = |z|e^{j\theta} = |z|\underline{\angle\theta} \tag{1.37}$$

where $|z|$ is the magnitude of z , θ is its phase angle,and $\underline{\angle\theta}$

is a useful shorthand representation for $e^{j\theta}$. Applying **Euler's identity**,

$$e^{j\theta} = \cos \theta + j \sin \theta, \tag{1.38}$$

we can convert z from polar form, as in Eq. (1.37), into rectangular form,

$$z = |z|e^{j\theta} = |z| \cos \theta + j|z| \sin \theta. \tag{1.39}$$

This leads to the relations

$$x = |z| \cos \theta, \quad y = |z| \sin \theta, \tag{1.40}$$

$$|z| = \sqrt[+]{x^2 + y^2}, \quad \theta = \tan^{-1}(y/x). \tag{1.41}$$

The two forms are illustrated graphically in **Fig. 1-18**. When using Eq. (1.41), care should be taken to ensure that θ is in the proper quadrant. Also note that, since $|z|$ is a positive quantity, only the positive root in Eq. (1.41) is applicable. This is denoted by the $+$ sign above the square-root sign.

The **complex conjugate** of z , denoted with a star superscript (or asterisk), is obtained by replacing j (wherever it appears) with $-j$, so that

$$z^* = (x + jy)^* = x - jy = |z|e^{-j\theta} = |z|\underline{\angle-\theta}. \tag{1.42}$$

The magnitude $|z|$ is equal to the positive square root of the product of z and its complex conjugate:

$$|z| = \sqrt[+]{zz^*}. \tag{1.43}$$

电磁波谱与1-6复数复习

图1-16见左侧原书：电磁波谱及大气透射窗口。下列中文图例与原图逐项对应。

图中：Gamma rays（伽马射线，癌症治疗）；X-rays（X射线，医学诊断）；Ultraviolet（紫外线，杀菌）；Visible（可见光）；Infrared（红外线，加热、夜视）；Radio spectrum（无线电波谱，通信、雷达、广播电视及射电天文）。Atmospheric opacity为大气不透明度；Optical、Infrared、Radio window分别为光学、红外、无线电窗口；Ionosphere opaque表示电离层不透明。下方刻度分别为波长（m）、频率（Hz）。

概念题1-9：电磁波具有哪三项基本性质？

概念题1-10：原书采用的微波频率范围是什么？

概念题1-11：可见光波长范围是多少？红外波段有哪些应用？

1-6 复数复习

任意复数可写为直角坐标形式 $z = x + jy$ ； x 、 y 分别是实部和虚部， $j^2 = -1$ ：

$$z = x + jy, \tag{1.35} \quad x = \Re(z), \quad y = \Im(z). \tag{1.36}$$

也可写为极坐标形式； $|z|$ 为模， θ 为辐角， $\angle\theta$ 为 $e^{j\theta}$ 的简写。利用欧拉公式，可在两种形式之间转换：

$$z = |z|e^{j\theta} = |z|\underline{\angle\theta} \tag{1.37} \quad e^{j\theta} = \cos \theta + j \sin \theta, \tag{1.38}$$

$$z = |z|e^{j\theta} = |z| \cos \theta + j|z| \sin \theta. \tag{1.39}$$

由此得到直角坐标与极坐标之间的关系：

$$x = |z| \cos \theta, \quad y = |z| \sin \theta, \tag{1.40}$$

$$|z| = \sqrt[+]{x^2 + y^2}, \quad \theta = \tan^{-1}(y/x). \tag{1.41}$$

如图1-18，求 θ 时须根据 x 、 y 的符号确定象限，不能只取 $\arctan(y/x)$ 的主值。 $|z|$ 为非负实数，应取平方根的非负值。

复共轭 z^* 是在 z 的直角坐标表达式中把 j 换成 $-j$ ；其模不变，辐角变号：

$$z^* = (x + jy)^* = x - jy = |z|e^{-j\theta} = |z|\underline{\angle-\theta}. \tag{1.42}$$

复数的模还可由 z 与其共轭的乘积求得：

$$|z| = \sqrt[+]{zz^*}. \tag{1.43}$$

数学校注： $z = 0$ 时 $|z| = 0$ ，辐角无定义；非零复数的辐角以 2π 为周期。计算可用atan2(y,x)正确处理象限。

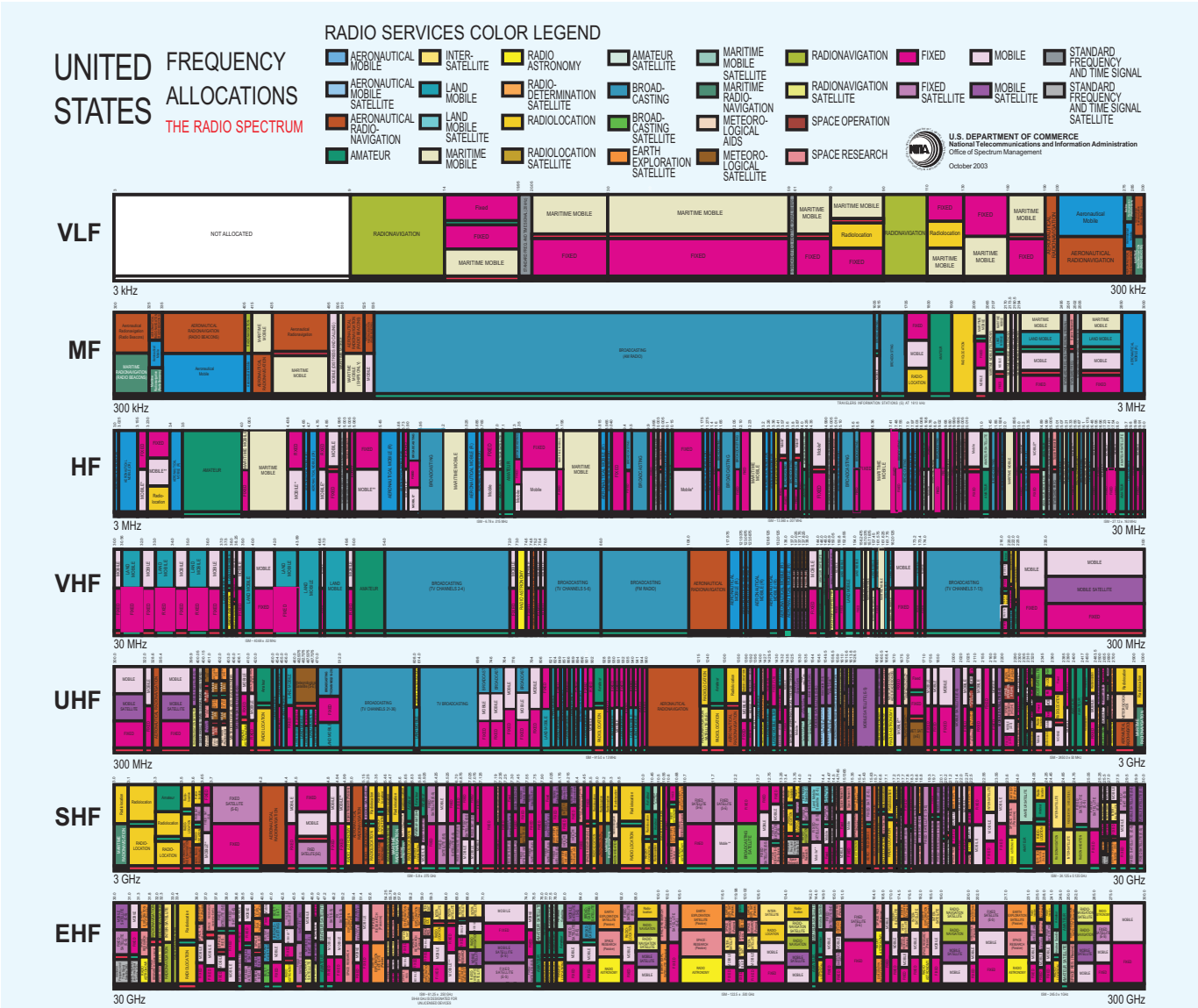


Figure 1-17 Individual bands of the radio spectrum and their primary allocations in the U.S. [See expandable version on book website: em8e.eecs.umich.edu.]

图1-17：无线电频谱图的中文对照

左侧保留原书完整频谱分配图，其来源为美国商务部国家电信和信息管理局（NTIA），日期为2003年10月。它说明当时美国无线电频谱的业务划分；原书提示可在配套网站查看放大版。

常用频段名称（对应原书命名）

缩写	中文名称	频率范围
VLF	甚低频	3—30 kHz
LF	低频	30—300 kHz
MF	中频	300 kHz—3 MHz
HF	高频	3—30 MHz
VHF	甚高频	30—300 MHz
UHF	特高频	300 MHz—3 GHz
SHF	超高频	3—30 GHz
EHF	极高频	30—300 GHz

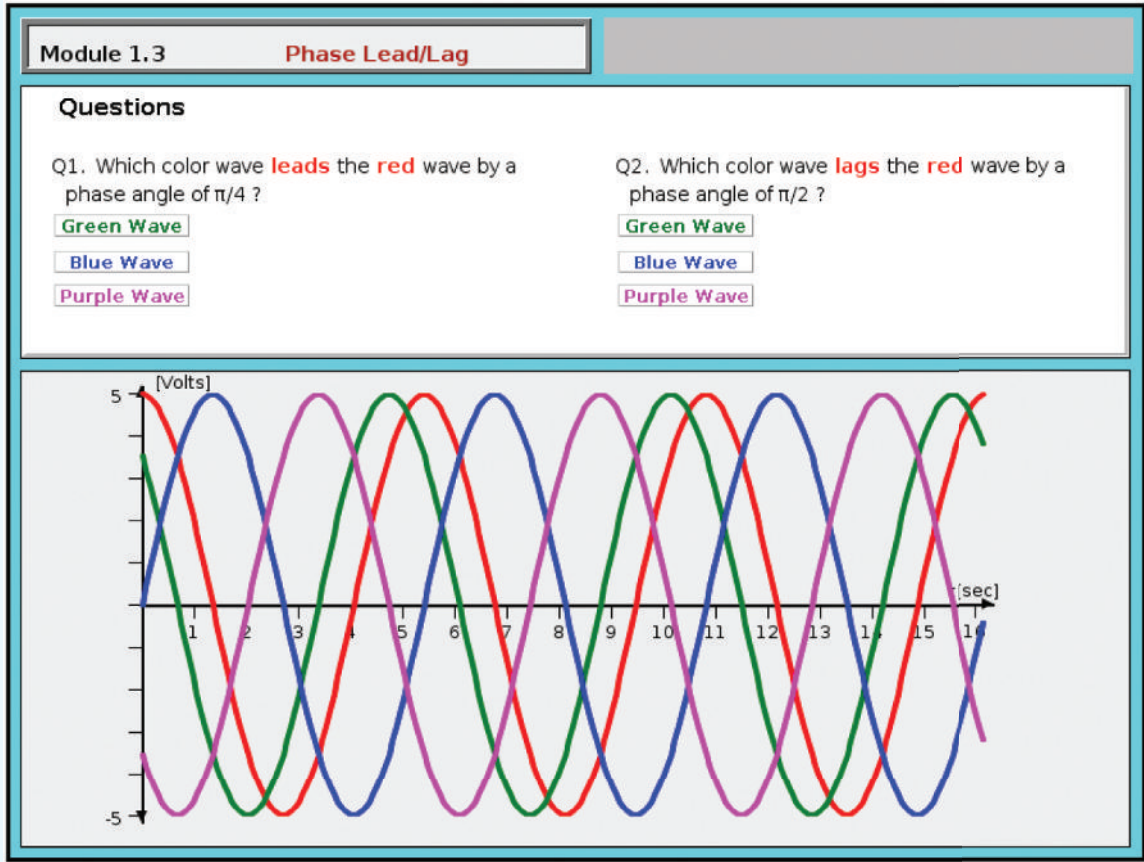
图中业务与标注词汇

英文	中文	英文	中文
Aeronautical mobile	航空移动	Aeronautical radionavigation	航空无线电导航
Amateur	业余	Broadcasting	广播
Earth exploration satellite	地球探测卫星	Fixed	固定
Inter-satellite	卫星间	Land mobile	陆地移动
Maritime mobile	水上移动	Maritime radionavigation	水上无线电导航
Meteorological aids	气象辅助	Meteorological satellite	气象卫星
Mobile	移动	Radio astronomy	射电天文
Radiodetermination satellite	无线电测定卫星	Radiolocation	无线电定位
Radionavigation	无线电导航	Space operation	空间操作
Space research	空间研究	Standard frequency and time signal	标准频率和时间信号
Satellite	卫星	Not allocated	未划分
Distress and calling	遇险与呼叫	Radio beacons	无线电信标
Telemetry / telemetering	遥测	Deep space	深空
Active / passive	有源 / 无源	E-S / S-E / S-S	地对空 / 空对地 / 空对空

复合业务名称按词组合理解，例如Fixed satellite为固定卫星业务，Broadcasting satellite为广播卫星业务，Aeronautical mobile satellite为航空移动卫星业务。图中的频率数字及业务分配位置保留在左侧原图，避免重绘误改。

年代校注：该图是2003年的美国频谱划分示意，不是现行规定，也不能作为其他国家的频率使用许可依据。本页补充完整图例词汇，未将历史分配表改写为当前分配表。

Module 1.3 Phase Lead/Lag Examine sinusoidal waveforms with different values of the reference phase constant ϕ_0 .



We now highlight some of the properties of complex algebra that will be encountered in future chapters.

Equality: If two complex numbers z_1 and z_2 are given by

$$z_1 = x_1 + jy_1 = |z_1|e^{j\theta_1}, \tag{1.44}$$

$$z_2 = x_2 + jy_2 = |z_2|e^{j\theta_2}, \tag{1.45}$$

then $z_1 = z_2$ if and only if $x_1 = x_2$ and $y_1 = y_2$ or, equivalently, $|z_1| = |z_2|$ and $\theta_1 = \theta_2$.

Addition:

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2). \tag{1.46}$$

Multiplication:

$$\begin{aligned} z_1 z_2 &= (x_1 + jy_1)(x_2 + jy_2) \\ &= (x_1 x_2 - y_1 y_2) + j(x_1 y_2 + x_2 y_1), \end{aligned} \tag{1.47a}$$

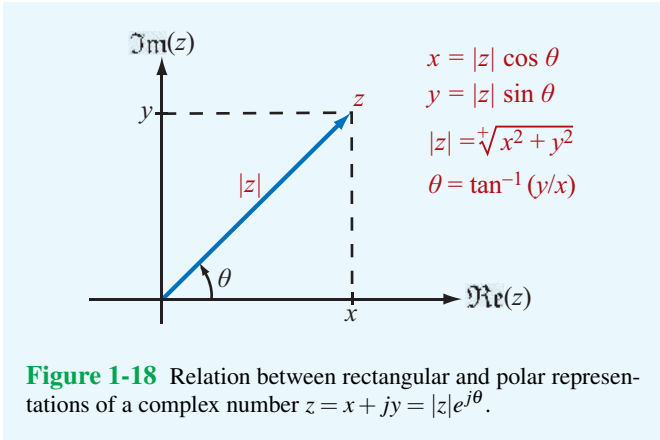
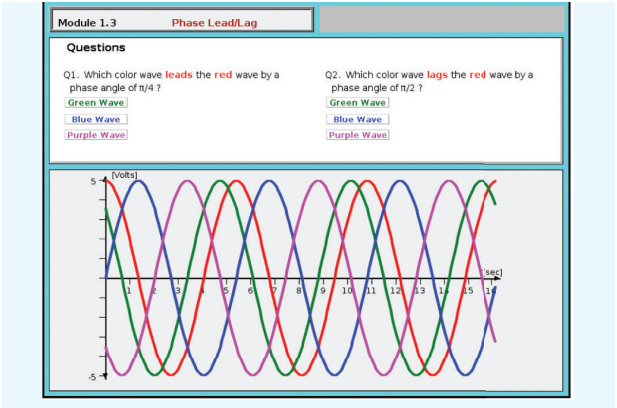


Figure 1-18 Relation between rectangular and polar representations of a complex number $z = x + jy = |z|e^{j\theta}$.

模块1.3与复数代数

模块1.3 相位超前 / 滞后：观察参考相位常数 ϕ_0 不同时的正弦波形。



原书模块1.3：相位差对时间波形的影响。

下面列出后续章节将使用的复数运算性质。设：

$$z_1 = x_1 + jy_1 = |z_1|e^{j\theta_1}, \tag{1.44}$$

$$z_2 = x_2 + jy_2 = |z_2|e^{j\theta_2}, \tag{1.45}$$

相等： $z_1 = z_2$ 当且仅当实部相等、虚部也相等。对于非零复数，等价条件是模相等，辐角在模 2π 意义下相等。

加法：实部与实部相加，虚部与虚部相加。

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2). \tag{1.46}$$

乘法：按代数展开，并利用 $j^2 = -1$ 。

$$\begin{aligned} z_1 z_2 &= (x_1 + jy_1)(x_2 + jy_2) \\ &= (x_1 x_2 - y_1 y_2) + j(x_1 y_2 + x_2 y_1), \end{aligned} \tag{1.47a}$$

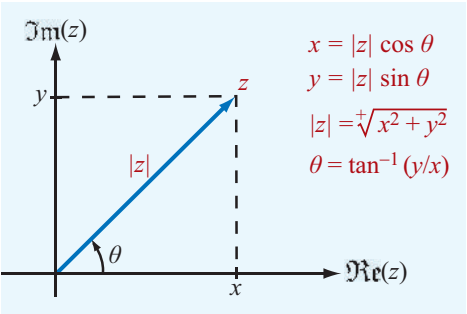


图1-18 复数z的直角坐标与极坐标表示：x为实部，y为虚部，|z|为模， θ 为辐角。

原书 “ $\theta_1 = \theta_2$ ” 的说法采用了相同辐角分支的约定；相差 $2n\pi$ 的辐角表示同一个非零复数。零复数的辐角没有定义。

or

$$\begin{aligned} z_1 z_2 &= |z_1|e^{j\theta_1} \cdot |z_2|e^{j\theta_2} \\ &= |z_1||z_2|e^{j(\theta_1+\theta_2)} \\ &= |z_1||z_2|[\cos(\theta_1+\theta_2) + j\sin(\theta_1+\theta_2)]. \end{aligned} \quad (1.47b)$$

Division: For $z_2 \neq 0$,

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{x_1 + jy_1}{x_2 + jy_2} = \frac{(x_1 + jy_1)}{(x_2 + jy_2)} \cdot \frac{(x_2 - jy_2)}{(x_2 - jy_2)} \\ &= \frac{(x_1x_2 + y_1y_2) + j(x_2y_1 - x_1y_2)}{x_2^2 + y_2^2}, \end{aligned} \quad (1.48a)$$

or

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{|z_1|e^{j\theta_1}}{|z_2|e^{j\theta_2}} = \frac{|z_1|}{|z_2|}e^{j(\theta_1-\theta_2)} \\ &= \frac{|z_1|}{|z_2|}[\cos(\theta_1-\theta_2) + j\sin(\theta_1-\theta_2)]. \end{aligned} \quad (1.48b)$$

Powers: For any positive integer n ,

$$\begin{aligned} z^n &= (|z|e^{j\theta})^n \\ &= |z|^n e^{jn\theta} = |z|^n (\cos n\theta + j\sin n\theta), \end{aligned} \quad (1.49)$$

$$z^{1/2} = \pm |z|^{1/2} e^{j\theta/2} = \pm |z|^{1/2} [\cos(\theta/2) + j\sin(\theta/2)]. \quad (1.50)$$

Useful Relations:

$$\begin{aligned} -1 &= e^{j\pi} = e^{-j\pi} = 1/\underline{180^\circ}, \\ j &= e^{j\pi/2} = 1/\underline{90^\circ}, \end{aligned} \quad (1.51)$$

$$-j = -e^{j\pi/2} = e^{-j\pi/2} = 1/\underline{-90^\circ}, \quad (1.52)$$

$$\sqrt{j} = (e^{j\pi/2})^{1/2} = \pm e^{j\pi/4} = \frac{\pm(1+j)}{\sqrt{2}}, \quad (1.53)$$

$$\sqrt{-j} = \pm e^{-j\pi/4} = \frac{\pm(1-j)}{\sqrt{2}}. \quad (1.54)$$

Example 1-3: Working with Complex Numbers

Given two complex numbers

$$V = 3 - j4, \quad I = -(2 + j3),$$

(a) express V and I in polar form, and find (b) VI , (c) VI^* , (d) V/I , and (e) $\sqrt{I}$.

Solution:

$$\begin{aligned} \text{(a)} \quad |V| &= \sqrt[3]{VV^*} \\ &= \sqrt[3]{(3-j4)(3+j4)} = \sqrt[3]{9+16} = 5, \end{aligned}$$

$$\theta_V = \tan^{-1}(-4/3) = -53.1^\circ,$$

$$V = |V|e^{j\theta_V} = 5e^{-j53.1^\circ} = 5/\underline{-53.1^\circ},$$

$$|I| = \sqrt[3]{2^2 + 3^2} = \sqrt[3]{13} = 3.61.$$

Since $I = (-2 - j3)$ is in the third quadrant in the complex plane (**Fig. 1-19**),

$$\theta_I = 180^\circ + \tan^{-1}\left(\frac{3}{2}\right) = 236.3^\circ,$$

$$I = 3.61/\underline{236.3^\circ}.$$

$$\begin{aligned} \text{(b)} \quad VI &= 5e^{-j53.1^\circ} \times 3.61e^{j236.3^\circ} \\ &= 18.03e^{j(236.3^\circ - 53.1^\circ)} = 18.03e^{j183.2^\circ}. \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad VI^* &= 5e^{-j53.1^\circ} \times 3.61e^{-j236.3^\circ} \\ &= 18.03e^{-j289.4^\circ} = 18.03e^{j70.6^\circ}. \end{aligned}$$

$$\text{(d)} \quad \frac{V}{I} = \frac{5e^{-j53.1^\circ}}{3.61e^{j236.3^\circ}} = 1.39e^{-j289.4^\circ} = 1.39e^{j70.6^\circ}.$$

$$\begin{aligned} \text{(e)} \quad \sqrt{I} &= \sqrt{3.61e^{j236.3^\circ}} \\ &= \pm\sqrt{3.61} e^{j236.3^\circ/2} = \pm 1.90e^{j118.15^\circ}. \end{aligned}$$

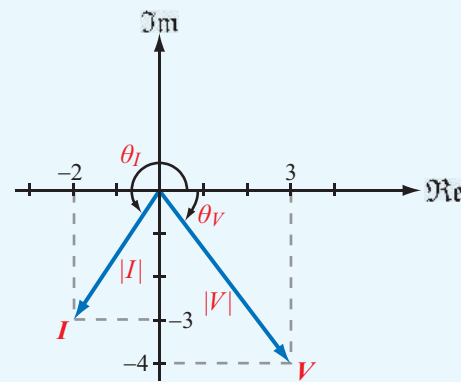


Figure 1-19 Complex numbers V and I in the complex plane (Example 1-3).

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或

$$\begin{aligned} z_1 z_2 &= |z_1|e^{j\theta_1} \cdot |z_2|e^{j\theta_2} \\ &= |z_1||z_2|e^{j(\theta_1+\theta_2)} \\ &= |z_1||z_2|[\cos(\theta_1+\theta_2) + j\sin(\theta_1+\theta_2)]. \end{aligned} \quad (1.47b)$$

除法：设 $z_2 \neq 0$,

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{x_1 + jy_1}{x_2 + jy_2} = \frac{(x_1 + jy_1)}{(x_2 + jy_2)} \cdot \frac{(x_2 - jy_2)}{(x_2 - jy_2)} \\ &= \frac{(x_1x_2 + y_1y_2) + j(x_2y_1 - x_1y_2)}{x_2^2 + y_2^2}, \end{aligned} \quad (1.48a)$$

或

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{|z_1|e^{j\theta_1}}{|z_2|e^{j\theta_2}} = \frac{|z_1|}{|z_2|}e^{j(\theta_1-\theta_2)} \\ &= \frac{|z_1|}{|z_2|}[\cos(\theta_1-\theta_2) + j\sin(\theta_1-\theta_2)]. \end{aligned} \quad (1.48b)$$

幂运算： n 为正整数，

$$\begin{aligned} z^n &= (|z|e^{j\theta})^n \\ &= |z|^n e^{jn\theta} = |z|^n (\cos n\theta + j\sin n\theta), \end{aligned} \quad (1.49)$$

$$z^{1/2} = \pm |z|^{1/2} e^{j\theta/2} = \pm |z|^{1/2} [\cos(\theta/2) + j\sin(\theta/2)]. \quad (1.50)$$

常用关系式：

$$\begin{aligned} -1 &= e^{j\pi} = e^{-j\pi} = 1/\underline{180^\circ}, \\ j &= e^{j\pi/2} = 1/\underline{90^\circ}, \end{aligned} \quad (1.51)$$

$$-j = -e^{j\pi/2} = e^{-j\pi/2} = 1/\underline{-90^\circ}, \quad (1.52)$$

$$\sqrt{j} = (e^{j\pi/2})^{1/2} = \pm e^{j\pi/4} = \frac{\pm(1+j)}{\sqrt{2}}, \quad (1.53)$$

$$\sqrt{-j} = \pm e^{-j\pi/4} = \frac{\pm(1-j)}{\sqrt{2}}. \quad (1.54)$$

例1-3 复数运算

已知两个复数：

$$V = 3 - j4, \quad I = -(2 + j3),$$

(a) 将 V 、 I 写成极坐标形式；求 (b) VI 、(c) VI^* 、(d) V/I 、(e) I 的两个平方根。

解：

$$\begin{aligned} \text{(a)} \quad |V| &= \sqrt[3]{VV^*} \\ &= \sqrt[3]{(3-j4)(3+j4)} = \sqrt[3]{9+16} = 5, \end{aligned}$$

$$\theta_V = \tan^{-1}(-4/3) = -53.1^\circ,$$

$$V = |V|e^{j\theta_V} = 5e^{-j53.1^\circ} = 5/\underline{-53.1^\circ},$$

$$|I| = \sqrt[3]{2^2 + 3^2} = \sqrt[3]{13} = 3.61.$$

$I = -2 - j3$ 位于复平面第三象限 (图1-19)，因此

$$\theta_I = 180^\circ + \tan^{-1}\left(\frac{3}{2}\right) = 236.3^\circ,$$

$$I = 3.61/\underline{236.3^\circ}.$$

$$\begin{aligned} \text{(b)} \quad VI &= 5e^{-j53.1^\circ} \times 3.61e^{j236.3^\circ} \\ &= 18.03e^{j(236.3^\circ - 53.1^\circ)} = 18.03e^{j183.2^\circ}. \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad VI^* &= 5e^{-j53.1^\circ} \times 3.61e^{-j236.3^\circ} \\ &= 18.03e^{-j289.4^\circ} = 18.03e^{j70.6^\circ}. \end{aligned}$$

$$\text{(d)} \quad \frac{V}{I} = \frac{5e^{-j53.1^\circ}}{3.61e^{j236.3^\circ}} = 1.39e^{-j289.4^\circ} = 1.39e^{j70.6^\circ}.$$

$$\begin{aligned} \text{(e)} \quad \sqrt{I} &= \sqrt{3.61e^{j236.3^\circ}} \\ &= \pm\sqrt{3.61} e^{j236.3^\circ/2} = \pm 1.90e^{j118.15^\circ}. \end{aligned}$$

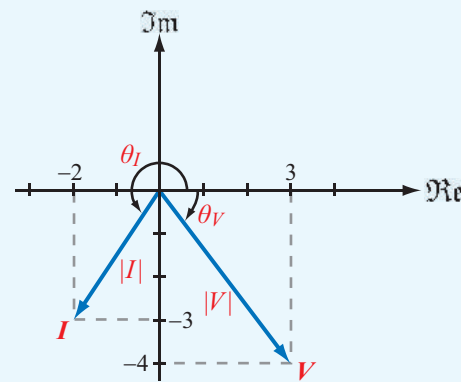


图1-19 复数 V 和 I 在复平面中的位置 (例1-3)。

校对： $VI = -18 - j$ ； $VI^* = 6 + j17$ ； $V/I = (6 + j17)/13$ 。平方根须保留±两支，不能把变量 I 翻译成汉字。

Technology Brief 2: Solar Cells

A **solar cell** is a photovoltaic device that converts solar energy into electricity. The conversion process relies on the **photovoltaic effect**, which was first reported by 19-year-old Edmund Bequerel in 1839 when he observed that a platinum electrode produced a small current if exposed to light. The photovoltaic effect is often confused with the **photoelectric effect**; they are interrelated, but not identical (Fig. TF2-1).

The photoelectric effect explains the mechanism responsible for why an electron is ejected by a material in consequence to a photon incident upon its surface (Fig. TF2-1(a)). For this to happen, the photon energy E (which is governed by its wavelength through $E = hc/\lambda$, with h being Planck's constant and c the velocity of light) has to exceed the binding energy with which the electron is held by the material. For his 1905 quantum-mechanical model of the photoelectric effect, Albert Einstein was awarded the 1921 Nobel Prize in physics.

Whereas a single material is sufficient for the photoelectric effect to occur, at least two adjoining materials with different electronic properties (to form a junction

that can support a voltage across it) are needed to establish a **photovoltaic current** through an external load (Fig. TF2-1(b)). Thus, the two effects are governed by the same quantum-mechanical rules associated with how photon energy can be used to liberate electrons away from their hosts, but the followup step of what happens to the liberated electrons is different in the two cases.

The PV Cell

Today's photovoltaic (PV) cells are made of semiconductor materials. The basic structure of a PV cell consists of a **p-n junction** connected to a load (Fig. TF2-2).

Typically, the n-type layer is made of silicon doped with a material that creates an abundance of negatively charged atoms, and the p-type layer also is made of silicon but doped with a different material that creates an abundance of holes (atoms with missing electrons). The combination of the two layers induces an electric field across the junction, so when an incident photon liberates an electron, the electron is swept under the influence of the electric field through the n-layer and out to the external circuit connected to the load.

The **conversion efficiency** of a PV cell depends on several factors, including the fraction of the incident light that gets absorbed by the semiconductor material, as opposed to getting reflected by the n-type front surface or transmitted through to the back conducting electrode. To minimize the reflected component, an antireflective coating usually is inserted between the upper glass cover and the n-type layer (Fig. TF2-2).

The PV cell shown in Fig. TF2-2 is called a **single-junction cell** because it contains only one p-n junction. The semiconductor material is characterized by a quantity called its **band gap energy**, which is the amount of energy needed to free an electron away from its host atom. For that to occur, the wavelength of the incident photon (which in turn defines its energy) has to be such that the photon's energy exceeds the band gap of the material. Solar energy extends over a broad spectrum, so only a fraction of the solar spectrum (photons with energies greater than the band gap) is absorbed by a single-junction material. To overcome this limitation, multiple p-n layers can be cascaded together to form a **multijunction PV device** (Fig. TF2-3). The cells usually are arranged such that the top cell has the highest band gap energy, thereby capturing the high-energy (short-wavelength) photons, followed by the cell with the next lower band gap, and so on.

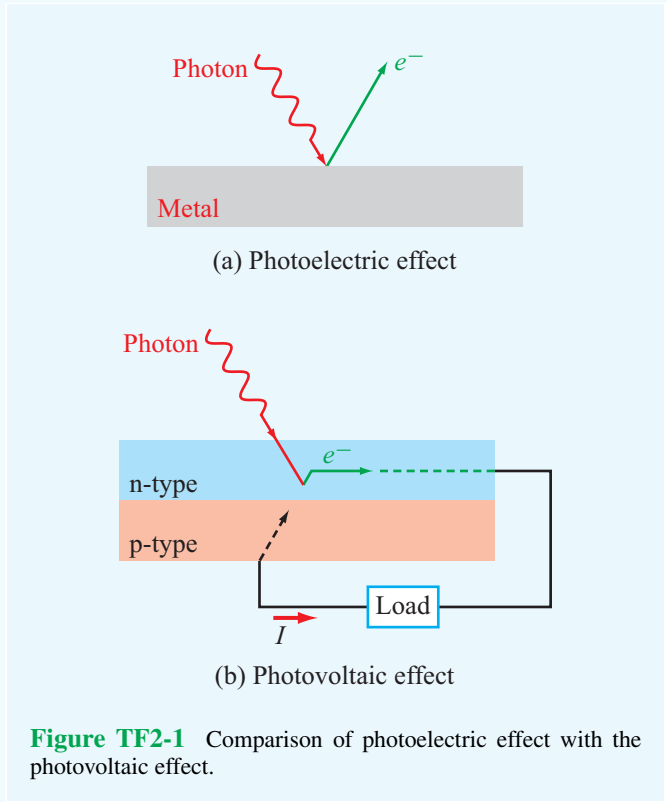


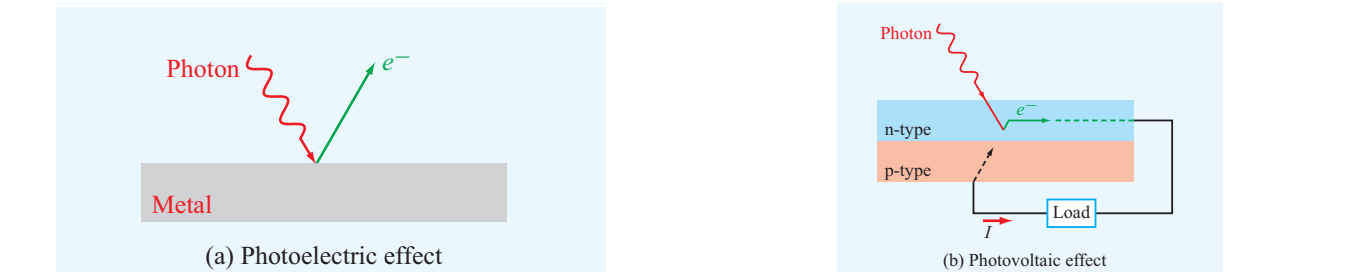
Figure TF2-1 Comparison of photoelectric effect with the photovoltaic effect.

技术短文2：太阳能电池

太阳能电池是一种把太阳能转换为电能的光伏器件。其转换过程利用光伏效应：1839年，19岁的贝克勒尔观察到铂电极受光照后产生微小电流，并报告了这一现象。光伏效应与光电效应有关，但二者并不相同（图TF2-1）。

本段所说的**光电效应**是外光电效应：入射光子使电子从材料表面逸出。光子能量 $E = hc/\lambda$ ，其中 h 为普朗克常数， c 为真空光速， λ 为真空波长。电子逸出要求光子能量达到材料的逸出功。爱因斯坦于1905年提出光电效应的光量子解释，并因对光电效应定律的发现获1921年诺贝尔物理学奖。

单一材料即可发生外光电效应。原书用两种电子特性不同、相互接触形成结的区域说明光伏效应：结两端能建立电压，接入负载后便可产生光生电流。两种效应都涉及光子能量对电子的激发，但激发后电子的运动过程不同。



图TF2-1(a) 外光电效应。Photon: 光子; Metal: 金属; e⁻: 电子。

图TF2-1(b) 光伏效应。n-type、p-type: n型、p型; Load: 负载; I: 电流。

光伏电池 (PV cell)

光伏电池使用半导体材料，典型结构是与外部负载相连的p-n结（图TF2-2）。在硅中掺入施主杂质可形成以电子为多数载流子的n型区；掺入受主杂质可形成以空穴为多数载流子的p型区。两区接触后，结区形成内建电场。光吸收产生的电子和空穴被分离、收集，电子可经n型区的电极进入外部负载回路。

转换效率受多种因素影响，其中之一是入射光中被半导体吸收的比例，其余光可能在前表面反射，或透射至背电极。为减小反射，通常在玻璃盖板与n型层之间设置减反射膜（图TF2-2）。

图TF2-2只有一个p-n结，称为单结电池。半导体的带隙能量决定其带间光吸收的能量阈值。太阳光覆盖较宽光谱，低于带隙的光子不能通过通常的带间吸收产生电子-空穴对，因此单结电池不能利用整个太阳光谱。多结器件把不同带隙的子电池层叠组合（图TF2-3）：最高带隙的顶层吸收高能、短波光光子，随后各层以逐渐降低的带隙利用透过的较低能光子。

学术校注：原文把n型掺杂说成产生“大量带负电的原子”，并把空穴称为“缺少电子的原子”，均不准确。n型多数载流子是电子，空穴是价带中的电子空态；电离施主带正电，均匀半导体体区通常近似电中性。带隙是导带底与价带顶的能量差，不是孤立原子的电离能。“两种材料”也不一定是两种不同化学物质，同种硅的不同掺杂区即可形成p-n结。以上正文已按这些区别订正，依据链接见校对进度文件。

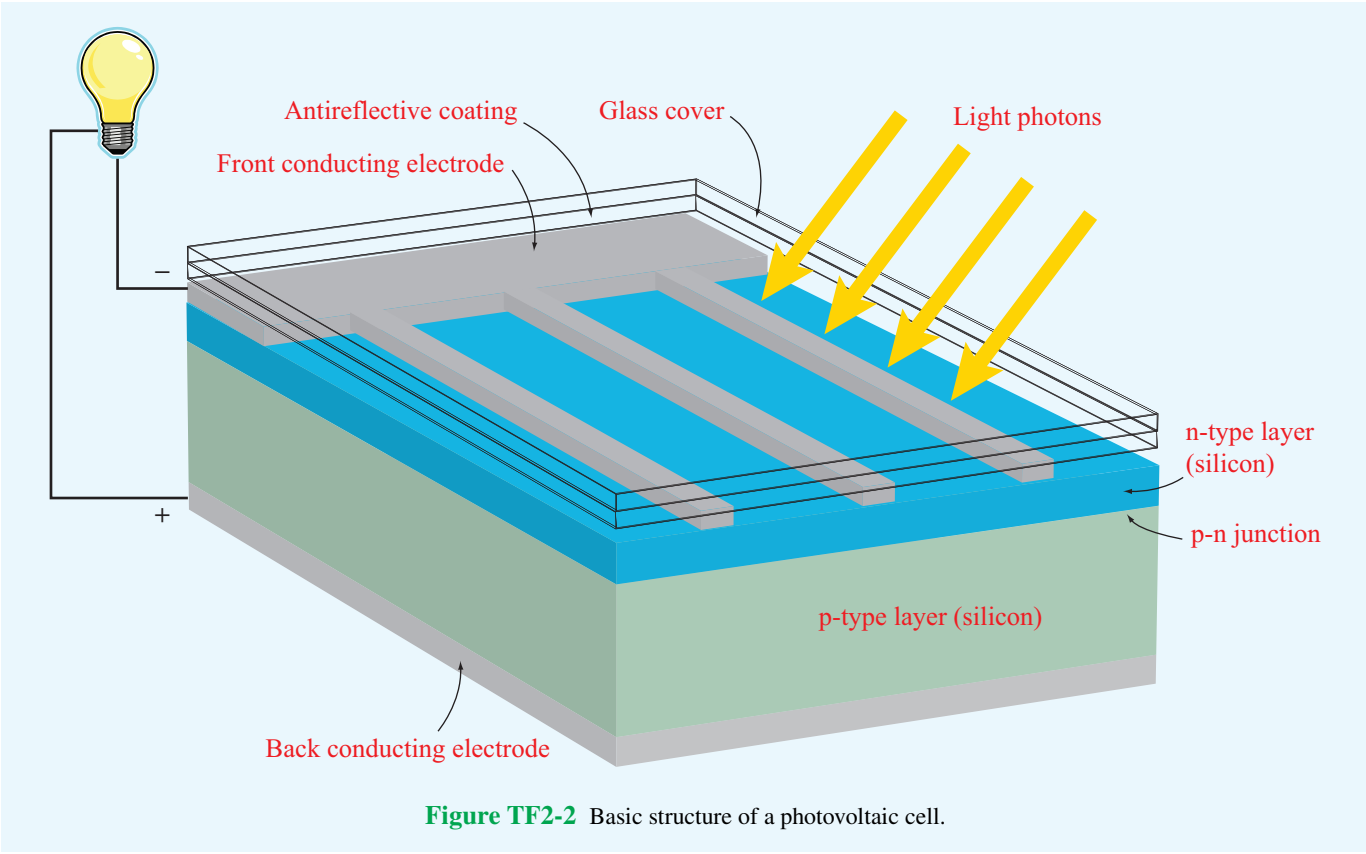


Figure TF2-2 Basic structure of a photovoltaic cell.

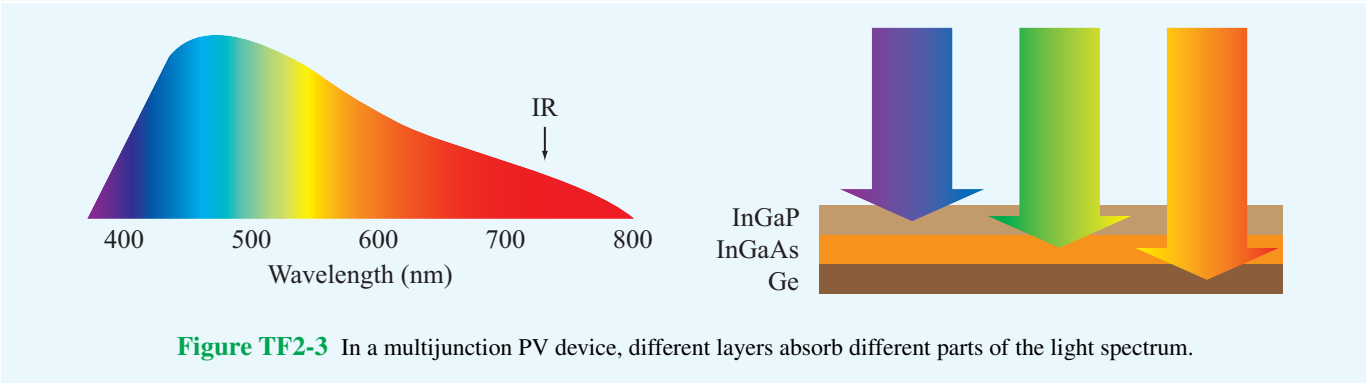
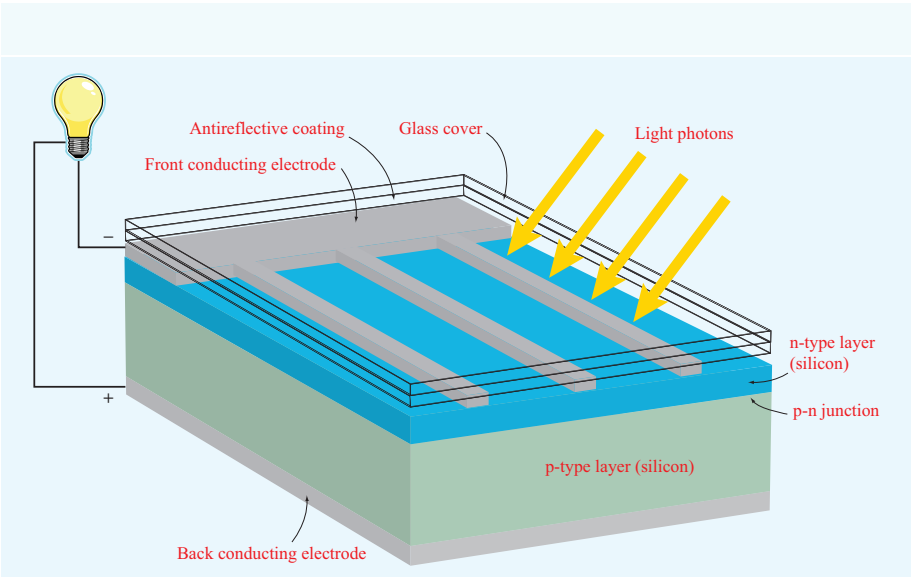


Figure TF2-3 In a multijunction PV device, different layers absorb different parts of the light spectrum.

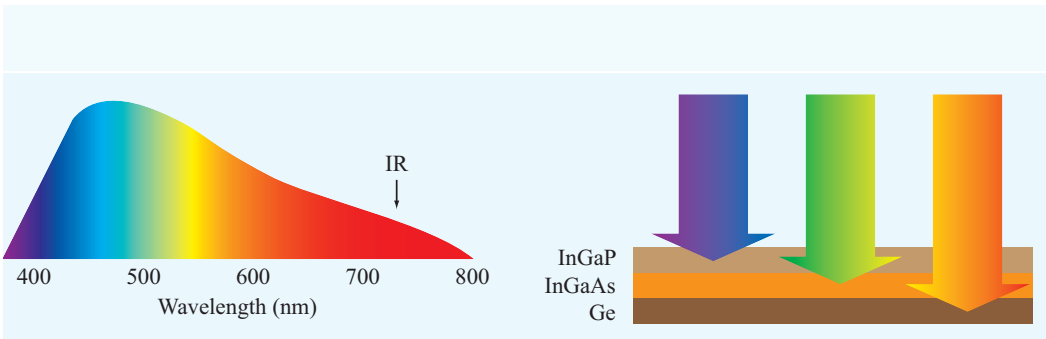
太阳能电池结构：图示与中文图例



图TF2-2 光伏电池的基本结构。原图的正负号、电极引线与层间关系保持不变。

图TF2-2 英中对照

英文标注	中文含义
Light photons	入射光子
Glass cover / Antireflective coating	玻璃盖板 / 减反射膜
Front / Back conducting electrode	前导电电极 / 背导电电极
n-type layer (silicon)	n型硅层
p-type layer (silicon)	p型硅层
p-n junction	p-n结



图TF2-3 多结光伏器件中，不同子电池吸收光谱的不同部分。

Wavelength (nm)：波长（纳米）；IR：红外。InGaP为磷化铟镓，InGaAs为砷化铟镓，Ge为锗。材料化学式中的In代表铟，不能把大写I误认作数字1或小写l。图中从短波到长波对应光子能量由高到低。

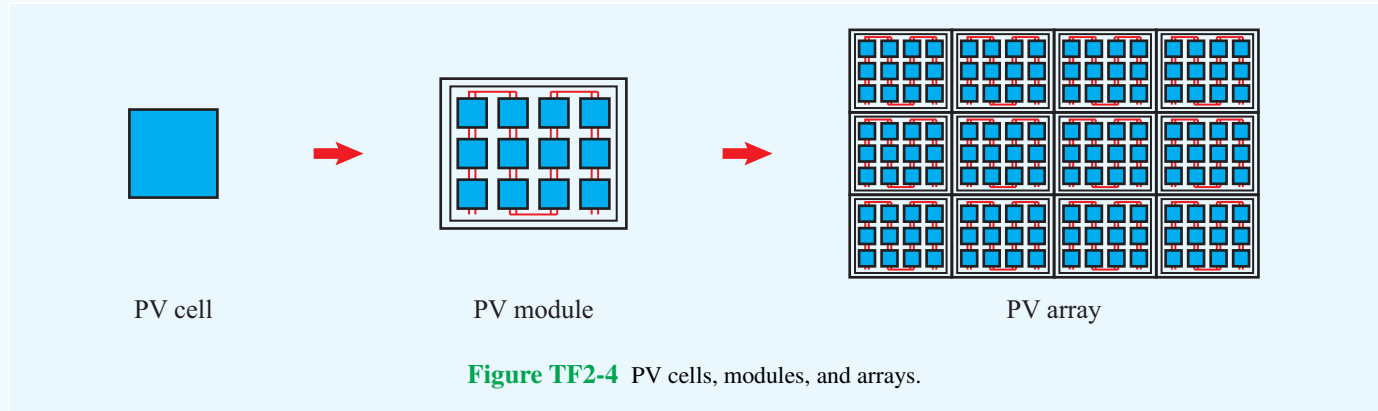


Figure TF2-4 PV cells, modules, and arrays.

► The multijunction technique offers an improvement in conversion efficiency of 2–4 times over that of the single-junction cell. However, the fabrication cost is significantly greater. ◀

Modules, Arrays, and Systems

A **photovoltaic module** consists of multiple PV cells connected together in order to supply electrical power at a specified voltage level, such as 12 or 24 V. The combination of multiple modules generates a **PV array** (Fig. TF2-4). The amount of generated power depends on the intensity of the intercepted sunlight, the total area of the module or array, and the conversion efficiencies of the individual cells. If the PV energy source is to serve multiple functions, it is integrated into an energy management system that includes a dc to ac current converter and batteries to store energy for later use (Fig. TF2-5).

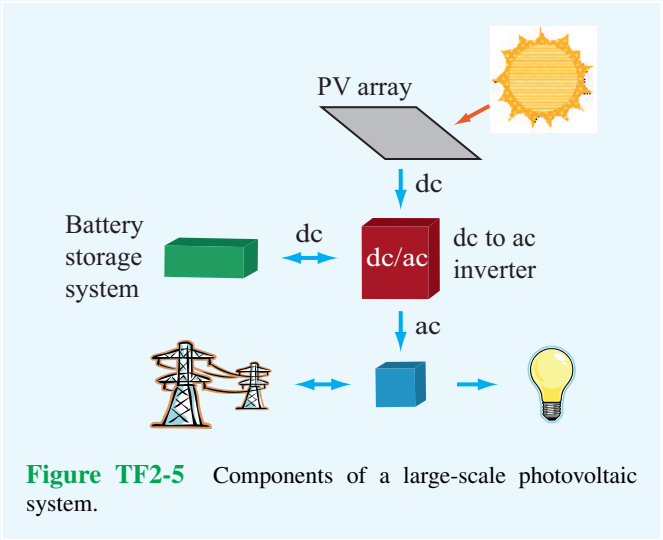
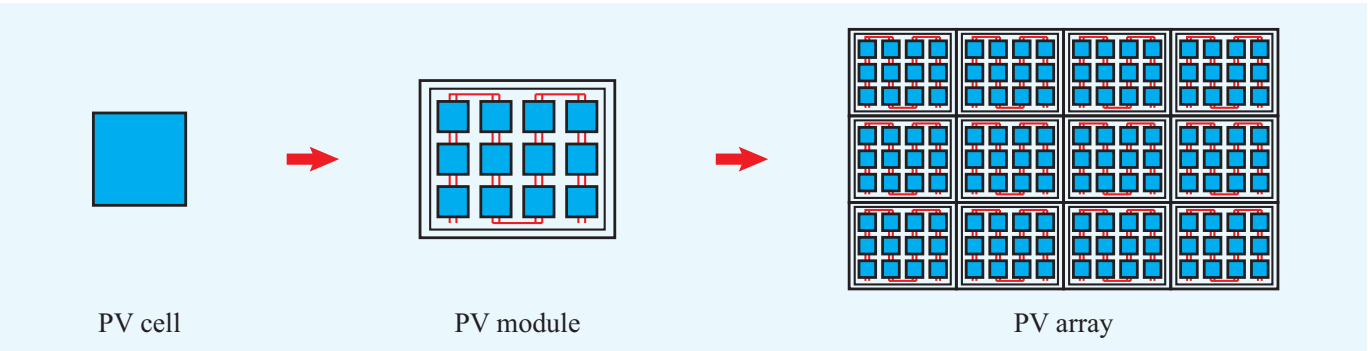


Figure TF2-5 Components of a large-scale photovoltaic system.

光伏组件、阵列与系统



图TF2-4 光伏电池（PV cell）、光伏组件（PV module）与光伏阵列（PV array）。

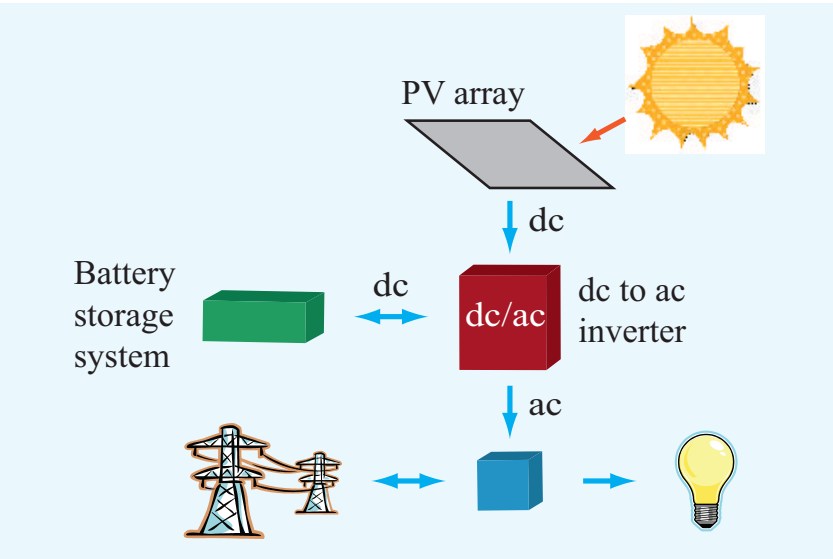
原书指出，多结技术可使转换效率达到单结电池的约2—4倍，但制造成本也显著增加。

比较口径校注：“达到约2—4倍”不等于“提高了2—4倍”。该数值是原书的概括性比较，未给出材料、光照和聚光条件，不能视为所有多结与单结器件间恒定的效率比例。

组件、阵列与系统

光伏组件由多个电池单元互连而成，以便在指定电压（例如12 V或24 V）下供电。多个组件组合成光伏阵列（图TF2-4）。其输出功率取决于接收到的太阳辐照强度、组件或阵列总面积，以及各电池的转换效率。这里讨论的是功率，不能与一段时间内累计产生的电能混淆。

若光伏电源需要承担多种供能功能，可将其集成到能源管理系统中。图TF2-5所示系统包括把直流转换为交流的逆变器，以及将电能储存起来供以后使用的蓄电池系统。



图TF2-5 大型光伏系统的组成。

图例：PV array为光伏阵列；dc to ac inverter为直流–交流逆变器；Battery storage system为蓄电池储能系统；dc、ac分别表示直流、交流。图示是一种含储能的系统配置，并不表示所有光伏系统都必须配置蓄电池。

Exercise 1-9: Express the following complex functions in polar form:

$$z_1 = (4 - j3)^2,$$

$$z_2 = (4 - j3)^{1/2}.$$

Answer: $z_1 = 25\angle -73.7^\circ$, $z_2 = \pm\sqrt{5}\angle -18.4^\circ$. (See [EM](#).)

Exercise 1-10: Show that $\sqrt{2j} = \pm(1 + j)$.

Answer: (See [EM](#).)

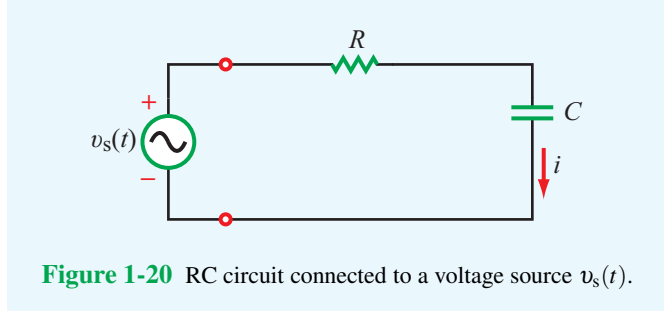


Figure 1-20 RC circuit connected to a voltage source $v_s(t)$.

gives the following loop equation:

$$Ri(t) + \frac{1}{C} \int i(t) dt = v_s(t). \quad (1.56)$$

(time domain)

Our objective is to obtain an expression for the current $i(t)$. We can do this by solving Eq. (1.56) in the time domain, which is somewhat cumbersome because the forcing function $v_s(t)$ is a sinusoid. Alternatively, we can take advantage of the phasor-domain solution technique as follows.

1-7.1 Solution Procedure

Step 1: Adopt a cosine reference

To establish a phase reference for all time-varying currents and voltages in the circuit, the forcing function is expressed as a cosine (if not already in that form). In the present example,

$$\begin{aligned} v_s(t) &= V_0 \sin(\omega t + \phi_0) \\ &= V_0 \cos\left(\frac{\pi}{2} - \omega t - \phi_0\right) \\ &= V_0 \cos\left(\omega t + \phi_0 - \frac{\pi}{2}\right), \end{aligned} \quad (1.57)$$

where we used the properties $\sin x = \cos(\pi/2 - x)$ and $\cos(-x) = \cos x$.

Step 2: Express time-dependent variables as phasors

Any cosinusoidally time-varying function $z(t)$ can be expressed as

$$z(t) = \Re \left[\tilde{Z} e^{j\omega t} \right], \quad (1.58)$$

where $\tilde{Z}$ is a time-independent function called the **phasor** of the **instantaneous** function $z(t)$. To distinguish instantaneous quantities from their phasor counterparts, a tilde ($\sim$) is added

1-7 相量复习

练习1-9: 将 $z_1 = (4 - j3)^2$ 及 $z_2 = (4 - j3)^{1/2}$ 写成极坐标形式。

答案: $z_1 = 25\angle(-73.7^\circ)$; z_2 的两根为 $\pm\sqrt{5}\angle(-18.4^\circ)$ 。详见配套电磁学资源 (EM)。

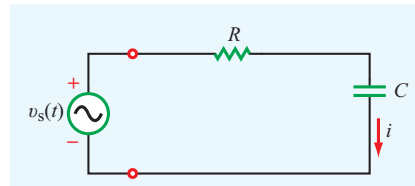
练习1-10: 证明 $2j$ 的两个平方根为 $\pm(1 + j)$ 。核算: $(\pm(1 + j))^2 = 2j$; 配套资源提供解答。

相量分析是处理线性系统正弦激励问题的数学工具。许多工程问题可写成线性积分微分方程; 用相量表示随时间正弦变化的变量, 可消去显式的正弦时间因子, 将求解过程简化。求得电压、电流等量的相量后, 再变换回时域。

对方波、脉冲序列等非正弦周期激励, 可先作傅里叶级数展开, 对每个正弦频率分量分别进行相量分析, 再由叠加原理合成时域响应。这样得到的受迫响应与直接在时域求解一致, 而步骤通常更简单。对单个脉冲等非周期激励, 可采用傅里叶积分并叠加各频率分量。

适用条件校注: 这里采用线性时不变系统及相应频率表示存在的条件; 单一相量描述给定频率的正弦稳态响应, 不能单独给出初始条件引起的暂态响应。不同频率须分别计算, 不能把它们作为同频相量直接相加。

图1-20的串联RC电路由正弦电压源驱动。 V_0 为振幅, ω 为角频率, ϕ_0 为参考相位。基尔霍夫电压定律给出式 (1.56) 的回路方程, 目标是求电流 $i(t)$ 。



$$v_s(t) = V_0 \sin(\omega t + \phi_0), \quad (1.55)$$

图1-20 串联RC电路; $v_s(t)$ 为电源电压。

$$Ri(t) + \frac{1}{C} \int i(t) dt = v_s(t). \quad (1.56)$$

(time domain)

式 (1.56) 为时域形式。直接求解包含正弦激励的积分微分方程较繁琐, 下面改用相量法。

1-7.1 求解步骤

步骤1: 统一采用余弦参考。为电路中各电压、电流建立共同参考, 先把激励改写为余弦形式:

$$\begin{aligned} v_s(t) &= V_0 \sin(\omega t + \phi_0) \\ &= V_0 \cos\left(\frac{\pi}{2} - \omega t - \phi_0\right) \\ &= V_0 \cos\left(\omega t + \phi_0 - \frac{\pi}{2}\right), \end{aligned} \quad (1.57)$$

这里使用 $\sin x = \cos(\pi/2 - x)$ 及 $\cos(-x) = \cos x$ 。

步骤2: 将时变量表示为相量。任意单频余弦函数 $z(t)$ 可写为:

$$z(t) = \Re \left[\tilde{Z} e^{j\omega t} \right], \quad (1.58)$$

式 (1.58) 中 $\Re$ 表示取实部, Z 上方加波浪号表示 $z(t)$ 的相量, 该相量不随时间变化。瞬时量与相量采用不同记号。(接下页)

over the letter representing a phasor. For the voltage $v_s(t)$ given by Eq. (1.57), its phasor equivalent $\tilde{V}_s$ is given by

$$\tilde{V}_s = V_0 e^{j(\phi_0 - \pi/2)}. \quad (1.59)$$

Thus, $\tilde{V}_s$ has the same amplitude as $v_s(t)$, and its exponential function is $j(\phi_0 - \pi/2)$, where $(\phi_0 - \pi/2)$ is the phase part of the cosine function in Eq. (1.57) without the ωt term. To verify that the expression for $\tilde{V}_s$ given by Eq. (1.59) is indeed the phasor equivalent of $v_s(t)$, we apply the right-hand side of Eq. (1.58):

$$\begin{aligned} \Re[\tilde{V}_s e^{j\omega t}] &= \Re[V_0 e^{j(\phi_0 - \pi/2)} \cdot e^{j\omega t}] \\ &= V_0 \cos(\omega t + \phi_0 - \pi/2), \end{aligned}$$

which is, indeed, the expression for $v_s(t)$ given by Eq. (1.57). The time-domain and phasor-domain correspondence can be expressed in the general form

$$v_s(t) = V_0 \cos(\omega t + \phi) \quad \longleftrightarrow \quad \tilde{V}_s = V_0 e^{j\phi}, \quad (1.60)$$

for any phase angle ϕ . Hence, ϕ may be a constant, such as $\phi_0 - \pi/2$, or it may be a function of one or more spatial variables, such as $\phi = \beta x$.

The phasor $\tilde{V}_s$, corresponding to the time function $v_s(t)$, contains amplitude and phase information but it is independent of the time variable t . Next we define the unknown variable $i(t)$ in terms of a phasor equivalent $\tilde{I}$ as

$$i(t) = \Re(\tilde{I} e^{j\omega t}). \quad (1.61)$$

If the equation we are trying to solve contains derivatives or integrals, we use the following two properties:

$$\frac{di}{dt} = \frac{d}{dt} [\Re(\tilde{I} e^{j\omega t})] = \Re \left[\frac{d}{dt} (\tilde{I} e^{j\omega t}) \right] = \Re[j\omega \tilde{I} e^{j\omega t}], \quad (1.62)$$

and

$$\begin{aligned} \int i \, dt &= \int \Re(\tilde{I} e^{j\omega t}) \, dt \\ &= \Re \left(\int \tilde{I} e^{j\omega t} \, dt \right) = \Re \left(\frac{\tilde{I}}{j\omega} e^{j\omega t} \right). \end{aligned} \quad (1.63)$$

► Thus, differentiation of the time function $i(t)$ with respect to time is equivalent to multiplication of its phasor $\tilde{I}$ by $j\omega$, and integration is equivalent to division by $j\omega$. ◀

Step 3: Recast the differential/integral equation in phasor form

Upon using the form of Eq. (1.58) to represent $i(t)$ and $v_s(t)$ in Eq. (1.56) in terms of their phasor equivalents, we have

$$R \Re(\tilde{I} e^{j\omega t}) + \frac{1}{C} \Re \left(\frac{\tilde{I}}{j\omega} e^{j\omega t} \right) = \Re(\tilde{V}_s e^{j\omega t}). \quad (1.64)$$

Combining all three terms under the same real-part ($\Re$) operator leads to

$$\Re \left\{ \left[\left(R + \frac{1}{j\omega C} \right) \tilde{I} - \tilde{V}_s \right] e^{j\omega t} \right\} = 0. \quad (1.65a)$$

Had we adopted a sine reference—instead of a cosine reference—to define sinusoidal functions, the preceding treatment would have led to the result

$$\Im \left\{ \left[\left(R + \frac{1}{j\omega C} \right) \tilde{I} - \tilde{V}_s \right] e^{j\omega t} \right\} = 0. \quad (1.65b)$$

Since both the real and imaginary parts of the expression inside the curly brackets are zero, the expression itself must be zero. Moreover, since $e^{j\omega t} \neq 0$, it follows that

$$\tilde{I} \left(R + \frac{1}{j\omega C} \right) = \tilde{V}_s \quad \textbf{(phasor domain)}. \quad (1.66)$$

The time factor $e^{j\omega t}$ has disappeared because it was contained in all three terms. Equation (1.66) is the phasor-domain equivalent of Eq. (1.56).

Step 4: Solve the phasor-domain equation

From Eq. (1.66), the phasor current $\tilde{I}$ is given by

$$\tilde{I} = \frac{\tilde{V}_s}{R + 1/(j\omega C)}. \quad (1.67)$$

Before we apply the next step, we need to convert the right-hand side of Eq. (1.67) into the form $I_0 e^{j\theta}$ with I_0 being a real quantity. Thus,

$$\begin{aligned} \tilde{I} &= V_0 e^{j(\phi_0 - \pi/2)} \left[\frac{j\omega C}{1 + j\omega RC} \right] \\ &= V_0 e^{j(\phi_0 - \pi/2)} \left[\frac{\omega C e^{j\pi/2}}{\sqrt{1 + \omega^2 R^2 C^2} e^{j\phi_1}} \right] \\ &= \frac{V_0 \omega C}{\sqrt{1 + \omega^2 R^2 C^2}} e^{j(\phi_0 - \phi_1)}, \end{aligned} \quad (1.68)$$

where we have used the identity $j = e^{j\pi/2}$. The phase angle associated with $(1 + j\omega RC)$ is $\phi_1 = \tan^{-1}(\omega RC)$ and lies in the first quadrant of the complex plane.

(承上页) 在相量字母上方加波浪号。式 (1.57) 电压的相量为：

$$\tilde{V}_s = V_0 e^{j(\phi_0 - \pi/2)}. \quad (1.59)$$

该相量的模等于电压振幅；其指数为 $j(\phi_0 - \pi/2)$ ，其中 $\phi_0 - \pi/2$ 是余弦参考相位，不含 ωt 。代入式 (1.58) 右端验证：

$$\begin{aligned} \Re[\tilde{V}_s e^{j\omega t}] &= \Re[V_0 e^{j(\phi_0 - \pi/2)} \cdot e^{j\omega t}] \\ &= V_0 \cos(\omega t + \phi_0 - \pi/2), \end{aligned}$$

所得结果正是式 (1.57)。对任意不随时间变化的相位 ϕ ，时域与相量域的一般对应关系为：

$$v_s(t) = V_0 \cos(\omega t + \phi) \quad \longleftrightarrow \quad \tilde{V}_s = V_0 e^{j\phi}, \quad (1.60)$$

ϕ 可以是常数（如 $\phi_0 - \pi/2$ ），也可以依赖空间坐标（如 βx ）。相量保留振幅和相位信息，不依赖时间 t 。未知电流 $i(t)$ 同样用其相量表示：

$$i(t) = \Re(\tilde{I} e^{j\omega t}). \quad (1.61)$$

若方程含时间导数或积分，可利用下列性质：

$$\frac{di}{dt} = \frac{d}{dt} [\Re(\tilde{I} e^{j\omega t})] = \Re \left[\frac{d}{dt} (\tilde{I} e^{j\omega t}) \right] = \Re[j\omega \tilde{I} e^{j\omega t}], \quad (1.62)$$

以及

$$\begin{aligned} \int i \, dt &= \int \Re(\tilde{I} e^{j\omega t}) \, dt \\ &= \Re \left(\int \tilde{I} e^{j\omega t} \, dt \right) = \Re \left(\frac{\tilde{I}}{j\omega} e^{j\omega t} \right). \end{aligned} \quad (1.63)$$

因此，对时间微分对应于相量乘 $j\omega$ ；对时间积分对应于相量除以 $j\omega$ 。校注：这里 $\omega \neq 0$ ，积分式只表示同频正弦分量，积分常数与初始条件须另行处理。

推导校注：不能把采用不同参考的两套相量当作同一组，从而断言实部、虚部同时为零。令方括号内系数为 $a + jb$ ，式 (1.65a) 即 $a \cos \omega t - b \sin \omega t = 0$ 对所有 t 成立，因此 $a = b = 0$ 。

步骤3：将微分 / 积分方程改写为相量方程

把式 (1.56) 的电流、电压代入式 (1.58) 形式，得：

$$R \Re(\tilde{I} e^{j\omega t}) + \frac{1}{C} \Re \left(\frac{\tilde{I}}{j\omega} e^{j\omega t} \right) = \Re(\tilde{V}_s e^{j\omega t}). \quad (1.64)$$

把三项合并在同一取实部算子 $\Re$ 之内，得：

$$\Re \left\{ \left[\left(R + \frac{1}{j\omega C} \right) \tilde{I} - \tilde{V}_s \right] e^{j\omega t} \right\} = 0. \quad (1.65a)$$

若采用正弦参考，并据此重新定义相量，可写成取虚部 $\Im$ 的形式：

$$\Im \left\{ \left[\left(R + \frac{1}{j\omega C} \right) \tilde{I} - \tilde{V}_s \right] e^{j\omega t} \right\} = 0. \quad (1.65b)$$

严格地说，式 (1.65a) 对所有 t 成立，故方括号内的复系数为零（ $\omega \neq 0$ ），于是：

$$\tilde{I} \left(R + \frac{1}{j\omega C} \right) = \tilde{V}_s \quad \textbf{(phasor domain)}. \quad (1.66)$$

各项具有相同时间因子 $e^{j\omega t}$ ，因此可将其消去。式 (1.66) 就是时域式 (1.56) 对应的相量域方程。

步骤4：求解相量域方程

由式 (1.66) 解得电流相量：

$$\tilde{I} = \frac{\tilde{V}_s}{R + 1/(j\omega C)}. \quad (1.67)$$

下一步先把结果写成 $I_0 e^{j\theta}$ 形式，其中 I_0 为非负实数：

$$\begin{aligned} \tilde{I} &= V_0 e^{j(\phi_0 - \pi/2)} \left[\frac{j\omega C}{1 + j\omega RC} \right] \\ &= V_0 e^{j(\phi_0 - \pi/2)} \left[\frac{\omega C e^{j\pi/2}}{\sqrt{1 + \omega^2 R^2 C^2} e^{j\phi_1}} \right] \\ &= \frac{V_0 \omega C}{\sqrt{1 + \omega^2 R^2 C^2}} e^{j(\phi_0 - \phi_1)}, \end{aligned} \quad (1.68)$$

这里使用 $j = e^{j\pi/2}$ 。对于 $R, C, \omega > 0$ ， $1 + j\omega RC$ 在第一象限，其辐角 $\phi_1 = \arctan(\omega RC)$ 。

Phasor Technique Solution Procedure

- (1) *Adopt a cosine reference* for all sinusoidally varying quantities, as in

$$x(t) = A \cos(\omega t + \phi_0).$$

- Coefficient A should be positive. If not, retain its positive magnitude and add or subtract π from ϕ_0 .
- If the time function is a sine instead of a cosine, convert it using Eq. (1.57).

- (2) *Convert time-dependent quantities into phasor equivalents:*

$$z(t) \longleftrightarrow \tilde{Z} \quad (\text{Table 1-5}).$$

- (3) *Cast equations in the phasor domain, and then solve them for the quantity of interest.*

- (4) If necessary, modify the solution $\tilde{Y}$ into the form

$$\tilde{Y} = B e^{j\theta},$$

where B is a positive, real quantity.

- (5) *Convert the phasor-domain solution to the time domain:*

$$y(t) = \Re\{B e^{j\theta} \cdot e^{j\omega t}\} = B \cos(\omega t + \theta)$$

Table 1-5 Time-domain sinusoidal functions $z(t)$ and their cosine-reference phasor-domain counterparts $\tilde{Z}$, where $z(t) = \Re[\tilde{Z}e^{j\omega t}]$.

$z(t)$	$\tilde{Z}$
$A \cos \omega t$	$\longleftrightarrow A$
$A \cos(\omega t + \phi_0)$	$\longleftrightarrow A e^{j\phi_0}$
$A \cos(\omega t + \beta x + \phi_0)$	$\longleftrightarrow A e^{j(\beta x + \phi_0)}$
$A e^{-\alpha x} \cos(\omega t + \beta x + \phi_0)$	$\longleftrightarrow A e^{-\alpha x} e^{j(\beta x + \phi_0)}$
$A \sin \omega t$	$\longleftrightarrow A e^{-j\pi/2}$
$A \sin(\omega t + \phi_0)$	$\longleftrightarrow A e^{j(\phi_0 - \pi/2)}$
$\frac{d}{dt}(z(t))$	$\longleftrightarrow j\omega \tilde{Z}$
$\frac{d}{dt}[A \cos(\omega t + \phi_0)]$	$\longleftrightarrow j\omega A e^{j\phi_0}$
$\int z(t) dt$	$\longleftrightarrow \frac{1}{j\omega} \tilde{Z}$
$\int A \sin(\omega t + \phi_0) dt$	$\longleftrightarrow \frac{1}{j\omega} A e^{j(\phi_0 - \pi/2)}$

Example 1-4: RL Circuit

The voltage source of the circuit shown in **Fig. 1-21** is given by

$$v_s(t) = 5 \sin(4 \times 10^4 t - 30^\circ) \quad (\text{V}). \quad (1.70)$$

Obtain an expression for the voltage across the inductor.

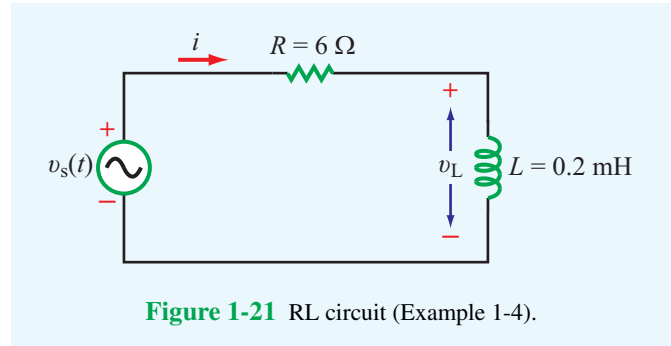


Figure 1-21 RL circuit (Example 1-4).

Solution: The voltage loop equation of the RL circuit is

$$Ri + L \frac{di}{dt} = v_s(t). \quad (1.71)$$

相量法步骤与例1-4：RL电路

相量法求解流程：

- (1) 统一余弦参考，写成 $x(t) = A \cos(\omega t + \phi_0)$ 。若 $A < 0$ ，改用 $|A|$ 并将相位增减 π ；正弦形式按式 (1.57) 转换。
- (2) 按表1-5把各时变量转换为相量。
- (3) 列出相量方程并求解所需量。
- (4) 把解写成模 B 与相位 θ 的极坐标形式， B 为非负实数。
- (5) 乘回 $e^{j\omega t}$ 并取实部，得到 $y(t) = B \cos(\omega t + \theta)$ 。

步骤5：变换回时域

按式 (1.61)，对式 (1.68) 的电流相量乘 $e^{j\omega t}$ 并取实部，即得串联RC电路的电流：

$$\begin{aligned} i(t) &= \Re[\tilde{I}e^{j\omega t}] = \Re\left[\frac{V_0 \omega C}{\sqrt{1 + \omega^2 R^2 C^2}} e^{j(\phi_0 - \phi_1)} e^{j\omega t}\right] \\ &= \frac{V_0 \omega C}{\sqrt{1 + \omega^2 R^2 C^2}} \cos(\omega t + \phi_0 - \phi_1). \quad (1.69) \end{aligned}$$

整个过程依次为：时变量转换为相量，求解电流相量，再变换回瞬时电流。表1-5列出常见函数与余弦参考相量的对应关系；相量均用字母上方的波浪号标识。

表1-5 时域函数 $z(t)$ 与相量的对应关系

$z(t)$	$\tilde{Z}$
$A \cos \omega t$	$\longleftrightarrow A$
$A \cos(\omega t + \phi_0)$	$\longleftrightarrow A e^{j\phi_0}$
$A \cos(\omega t + \beta x + \phi_0)$	$\longleftrightarrow A e^{j(\beta x + \phi_0)}$
$A e^{-\alpha x} \cos(\omega t + \beta x + \phi_0)$	$\longleftrightarrow A e^{-\alpha x} e^{j(\beta x + \phi_0)}$
$A \sin \omega t$	$\longleftrightarrow A e^{-j\pi/2}$
$A \sin(\omega t + \phi_0)$	$\longleftrightarrow A e^{j(\phi_0 - \pi/2)}$
$\frac{d}{dt}(z(t))$	$\longleftrightarrow j\omega \tilde{Z}$
$\frac{d}{dt}[A \cos(\omega t + \phi_0)]$	$\longleftrightarrow j\omega A e^{j\phi_0}$
$\int z(t) dt$	$\longleftrightarrow \frac{1}{j\omega} \tilde{Z}$
$\int A \sin(\omega t + \phi_0) dt$	$\longleftrightarrow \frac{1}{j\omega} A e^{j(\phi_0 - \pi/2)}$

左列为时域函数，右列为相量；微分对应乘 $j\omega$ ，积分对应除以 $j\omega$ ($\omega \neq 0$ ，积分常数另计)。

例1-4 RL电路

图1-21的电源电压如下。已知 $R = 6 \Omega$ ， $L = 0.2 \text{ mH}$ ，求电感两端的电压表达式。

$$v_s(t) = 5 \sin(4 \times 10^4 t - 30^\circ) \quad (\text{V}). \quad (1.70)$$

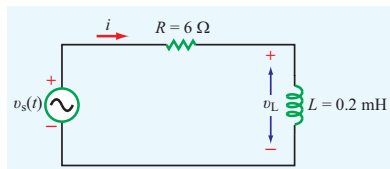


图1-21 串联RL电路，电感电压极性按原图。

$$Ri + L \frac{di}{dt} = v_s(t). \quad (1.71)$$

解：按图示电流方向与电压极性，由基尔霍夫电压定律得到式 (1.71) 的回路方程。（接下页）

例1-4续解与相量域中的行波

先将式（1.70）的电源电压改写为余弦参考，再把式（1.71）转换为相量方程：

$$\begin{aligned} v_s(t) &= 5 \sin(4 \times 10^4 t - 30^\circ) \\ &= 5 \cos(4 \times 10^4 t - 120^\circ) \quad (\text{V}). \end{aligned} \quad (1.72)$$

$$R\tilde{I} + j\omega L\tilde{I} = \tilde{V}_s. \quad (1.73)$$

角频率 $\omega = 4 \times 10^4$ rad/s。由表1-5，电源相量为 $5\angle(-120^\circ)$ V；回路阻抗 $R + j\omega L = 6 + j8 = 10\angle 53.1^\circ \Omega$ 。求电流相量：

$$\begin{aligned} \tilde{I} &= \frac{\tilde{V}_s}{R + j\omega L} \\ &= \frac{5e^{-j120^\circ}}{6 + j4 \times 10^4 \times 2 \times 10^{-4}} \\ &= \frac{5e^{-j120^\circ}}{6 + j8} = \frac{5e^{-j120^\circ}}{10e^{j53.1^\circ}} = 0.5e^{-j173.1^\circ} \quad (\text{A}). \end{aligned}$$

电感电压相量等于电流相量乘 $j\omega L$ ；再乘 $e^{j\omega t}$ 并取实部，即得瞬时电压：

$$\begin{aligned} \tilde{V}_L &= j\omega L\tilde{I} \\ &= j4 \times 10^4 \times 2 \times 10^{-4} \times 0.5e^{-j173.1^\circ} \\ &= 4e^{j(90^\circ - 173.1^\circ)} = 4e^{-j83.1^\circ} \quad (\text{V}), \end{aligned}$$

$$\begin{aligned} v_L(t) &= \Re \left[\tilde{V}_L e^{j\omega t} \right] \\ &= \Re \left[4e^{-j83.1^\circ} e^{j4 \times 10^4 t} \right] \\ &= 4 \cos(4 \times 10^4 t - 83.1^\circ) \quad (\text{V}). \end{aligned}$$

电感电压相量。对应的瞬时电压。

概念题1-12：相量法为什么有用？适用于何种情况？说明求解过程。

概念题1-13：激励为脉冲串等非正弦周期波形时，如何使用相量法？

练习1-11：串联RL电路的电源电压为 $v_s(t) = 150 \cos \omega t$ V。已知 $R = 400 \Omega$ ， $L = 3 \text{ mH}$ ， $\omega = 10^5$ rad/s，求（a）电流相量；（b）瞬时电流。

答案：（a） $150/(400 + j300) = 0.3\angle(-36.9^\circ)$ A；（b） $i(t) = 0.3 \cos(\omega t - 36.9^\circ)$ A。详见配套资源。

练习1-12：电压相量为 $j5$ V，求瞬时电压。

答案： $v(t) = 5 \cos(\omega t + \pi/2) = -5 \sin \omega t$ V。详见配套资源。

1-7.2 相量域中的行波

在表1-5第三项中令 $\phi_0 = 0$ ，得到：

$$A \cos(\omega t + \beta x) \quad \longleftrightarrow \quad Ae^{j\beta x}. \quad (1.74)$$

由式（1.31）的讨论， $A \cos(\omega t + \beta x)$ 沿 $-x$ 方向传播。在本书采用的 $e^{j\omega t}$ 时间因子、 $\beta > 0$ 约定下，无损介质中沿 $+x$ 方向传播的波，其相量为 $Ae^{-j\beta x}$ ；沿 $-x$ 方向传播的波，其相量为 $Ae^{j\beta x}$ 。空间指数中 x 项的符号与传播方向相反。

校核：例1-4电感电压振幅为4 V，余弦参考相位约 -83.13° ；练习1-11电流振幅0.3 A、相位约 -36.87° 。上述相位取到原书精度；相量法结果为正弦稳态解。

Before converting Eq. (1.71) into the phasor domain, we express Eq. (1.70) in terms of a cosine reference:

$$\begin{aligned} v_s(t) &= 5 \sin(4 \times 10^4 t - 30^\circ) \\ &= 5 \cos(4 \times 10^4 t - 120^\circ) \quad (\text{V}). \end{aligned} \quad (1.72)$$

The coefficient of t specifies the angular frequency as $\omega = 4 \times 10^4$ (rad/s). Per the second entry in **Table 1-5**, the voltage phasor corresponding to $v_s(t)$ is

$$\tilde{V}_s = 5e^{-j120^\circ} \quad (\text{V}),$$

and the phasor equation corresponding to Eq. (1.71) is

$$R\tilde{I} + j\omega L\tilde{I} = \tilde{V}_s. \quad (1.73)$$

Solving for the current phasor $\tilde{I}$, we have

$$\begin{aligned} \tilde{I} &= \frac{\tilde{V}_s}{R + j\omega L} \\ &= \frac{5e^{-j120^\circ}}{6 + j4 \times 10^4 \times 2 \times 10^{-4}} \\ &= \frac{5e^{-j120^\circ}}{6 + j8} = \frac{5e^{-j120^\circ}}{10e^{j53.1^\circ}} = 0.5e^{-j173.1^\circ} \quad (\text{A}). \end{aligned}$$

The voltage phasor across the inductor is related to $\tilde{I}$ by

$$\begin{aligned} \tilde{V}_L &= j\omega L\tilde{I} \\ &= j4 \times 10^4 \times 2 \times 10^{-4} \times 0.5e^{-j173.1^\circ} \\ &= 4e^{j(90^\circ - 173.1^\circ)} = 4e^{-j83.1^\circ} \quad (\text{V}), \end{aligned}$$

and the corresponding instantaneous voltage $v_L(t)$ is therefore

$$\begin{aligned} v_L(t) &= \Re \left[\tilde{V}_L e^{j\omega t} \right] \\ &= \Re \left[4e^{-j83.1^\circ} e^{j4 \times 10^4 t} \right] \\ &= 4 \cos(4 \times 10^4 t - 83.1^\circ) \quad (\text{V}). \end{aligned}$$

Concept Question 1-12: Why is the phasor technique useful? When is it used? Describe the process.

Concept Question 1-13: How is the phasor technique used when the forcing function is a nonsinusoidal periodic waveform, such as a train of pulses?

Exercise 1-11: A series RL circuit is connected to a voltage source given by $v_s(t) = 150 \cos \omega t$ (V). Find (a) the phasor current $\tilde{I}$ and (b) the instantaneous current $i(t)$ for $R = 400 \Omega$, $L = 3 \text{ mH}$, and $\omega = 10^5$ rad/s.

Answer: (a) $\tilde{I} = 150/(R + j\omega L) = 0.3\angle-36.9^\circ$ (A), (b) $i(t) = 0.3 \cos(\omega t - 36.9^\circ)$ (A). (See [EM](#).)

Exercise 1-12: A phasor voltage is given by $\tilde{V} = j5$ V. Find $v(t)$.

Answer: $v(t) = 5 \cos(\omega t + \pi/2) = -5 \sin \omega t$ (V). (See [EM](#).)

1-7.2 Traveling Waves in the Phasor Domain

According to **Table 1-5**, if we set $\phi_0 = 0$, its third entry becomes

$$A \cos(\omega t + \beta x) \quad \longleftrightarrow \quad Ae^{j\beta x}. \quad (1.74)$$

From the discussion associated with Eq. (1.31), we concluded that $A \cos(\omega t + \beta x)$ describes a wave traveling in the negative x direction.

► In the phasor domain, a wave of amplitude A traveling in the positive x direction in a lossless medium with a phase constant β is given by the negative exponential $Ae^{-j\beta x}$. Conversely, a wave traveling in the negative x direction is given by $Ae^{j\beta x}$. Thus, the sign of x in the exponential is opposite to the direction of travel. ◀

Chapter 1 Summary

Concepts

- Electromagnetics is the study of electric and magnetic phenomena and their engineering applications.
- The International System of Units consists of the six fundamental dimensions listed in **Table 1-1**. The units of all other physical quantities can be expressed in terms of the six fundamental units.
- The four fundamental forces of nature are nuclear, weak-interaction, electromagnetic, and gravitational forces.
- The source of the electric field quantities **E** and **D** is the electric charge q . In a material, **E** and **D** are related by $\mathbf{D} = \epsilon \mathbf{E}$, where ϵ is the electrical permittivity of the material. In free space, $\epsilon = \epsilon_0 \approx (1/36\pi) \times 10^{-9}$ (F/m).
- The source of the magnetic field quantities **B** and **H** is the electric current I . In a material, **B** and **H** are related by $\mathbf{B} = \mu \mathbf{H}$, where μ is the magnetic permeability of the medium. In free space, $\mu = \mu_0 = 4\pi \times 10^{-7}$ (H/m).
- Electromagnetics consists of three branches:
 - electrostatics, which pertains to stationary charges,
 - magnetostatics, which pertains to dc currents, and
 - electrodynamics, which pertains to time-varying currents.
- A traveling wave is characterized by a spatial wavelength λ , a time period T , and a phase velocity $u_p = \lambda/T$.
- An electromagnetic (EM) wave consists of oscillating electric and magnetic field intensities and travels in free space at the velocity of light: $c = 1/\sqrt{\epsilon_0\mu_0}$. The EM spectrum encompasses gamma rays, X-rays, visible light, infrared waves, and radio waves.
- Phasor analysis is a useful mathematical tool for solving problems involving time-periodic sources.

Mathematical and Physical Models

Electric field due to charge q in free space

$$\mathbf{E} = \hat{\mathbf{R}} \frac{q}{4\pi\epsilon_0 R^2}$$

Magnetic field due to current I in free space

$$\mathbf{B} = \hat{\boldsymbol{\phi}} \frac{\mu_0 I}{2\pi r}$$

Plane wave $y(x,t) = Ae^{-\alpha x} \cos(\omega t - \beta x + \phi_0)$

- $\alpha = 0$ in lossless medium
- phase velocity $u_p = f\lambda = \frac{\omega}{\beta}$
- $\omega = 2\pi f$; $\beta = 2\pi/\lambda$
- $\phi_0 =$ phase reference

Complex numbers

- Euler’s identity
 $e^{j\theta} = \cos \theta + j \sin \theta$
- Rectangular-polar relations
 $x = |z| \cos \theta, \quad y = |z| \sin \theta,$
 $|z| = \sqrt{x^2 + y^2}, \quad \theta = \tan^{-1}(y/x)$

Phasor-domain equivalents

See **Table 1-5**

following:

- (a) $p(x,t)$ versus x at $t = 0$
- (b) $p(x,t)$ versus t at $x = 0$

Be sure to use appropriate scales for x and t so that each of your plots covers at least two cycles.

***1.3** A 2 kHz sound wave traveling in the x direction in air was observed to have a differential pressure $p(x,t) = 10$ N/m² at $x = 0$ and $t = 50 \mu\text{s}$. If the reference phase of $p(x,t)$ is 36°, find a complete expression for $p(x,t)$. The velocity of sound in air is 330 m/s.

第一章总结

概念

- 电磁学研究电、磁现象及其工程应用。
- 国际单位制（SI）有七个基本单位。表1-1列出本书常用的六种基本量，未列物质的量及其单位摩尔。[校注1]
- 自然界的四种基本相互作用为强相互作用、弱相互作用、电磁相互作用和引力相互作用。
- 电荷 q 可产生电场。在本书采用的线性各向同性介质中， $\mathbf{D} = \epsilon \mathbf{E}$ ， ϵ 为介电率；真空中 $\epsilon_0 \approx (1/36\pi) \times 10^{-9}$ F/m。
- 电流 I 可产生磁场；在线性各向同性介质中， $\mathbf{B} = \mu \mathbf{H}$ ， μ 为磁导率。真空中 $\mu_0 \approx 4\pi \times 10^{-7}$ H/m。[校注1]
- 本书将电磁学分为静电学（静止电荷）、静磁学（恒定电流）和电动力学（时变电磁现象）。
- 行波由空间波长 λ 、时间周期 T 和相速度 $u_p = \lambda/T$ 描述。
- 电磁波由随时间变化的电场与磁场组成，在真空中以光速 $c = 1/\sqrt{\epsilon_0\mu_0}$ 传播。电磁频谱包括伽马射线、X射线、紫外线、可见光、红外线、微波和无线电波等。
- 相量分析适用于同一频率的正弦稳态问题。非正弦周期信号可先作傅里叶分解，再分别分析各频率分量。

数学与物理模型

真空中点电荷 q 产生的电场

$$\mathbf{E} = \hat{\mathbf{R}} \frac{q}{4\pi\epsilon_0 R^2}$$

真空中无限长直导线电流 I 产生的磁场

$$\mathbf{B} = \hat{\boldsymbol{\phi}} \frac{\mu_0 I}{2\pi r}$$

平面波 $y(x,t) = Ae^{-\alpha x} \cos(\omega t - \beta x + \phi_0)$

- $\alpha = 0$ 在无损介质中
- 相速度 $u_p = f\lambda = \frac{\omega}{\beta}$
- $\omega = 2\pi f$; $\beta = 2\pi/\lambda$
- $\phi_0 =$ 参考相位

复数

- 欧拉恒等式
 $e^{j\theta} = \cos \theta + j \sin \theta$
- 直角形式与极坐标形式的关系
 $x = |z| \cos \theta, \quad y = |z| \sin \theta,$
 $|z| = \sqrt{x^2 + y^2}, \quad \theta = \tan^{-1}(y/x)$

相量域对应关系

见表1-5

辐角 θ 须结合 x 、 y 的象限确定； $z=0$ 时辐角无定义。

问题

第 1-4 节：行波

***1.1** 沿弦传播的谐波由每分钟完成 180 次振动的振荡器产生。如果观察到给定的波峰或最大值在 10 秒内传播 300 厘米，那么波长是多少？

1.2 对于例 1-1 中描述的压力波，绘制

* 标记的习题在附录 E 提供答案。

下列的：

- (a) $p(x,t)$ 与 $t = 0$ 处的 x
- (b) $p(x,t)$ 与 $x = 0$ 处的 t

请务必对 x 和 t 使用适当的比例，以便每个图至少涵盖两个周期。

***1.3** 观察到在空气中沿 x 方向传播的 2 kHz 声波在 $x = 0$ 和 $t = 50 \mu\text{s}$ 处具有压差 $p(x,t) = 10$ N/m²。如果 $p(x,t)$ 的参考相位是 36°，则找到 $p(x,t)$ 的完整表达式。空气中的声速为 330 m/s。

校注1：原书“六个基本单位”不完整。另自2019年SI修订后， μ_0 不再由定义固定为 $4\pi \times 10^{-7}$ H/m；此处为工程近似。依据：BIPM《SI手册》第9版及第26届CGPM第1号决议。

* Answer(s) available in Appendix E.

Important Terms			Provide definitions or explain the meaning of the following terms:
angular velocity ω	EM spectrum	phase	
attenuation constant α	Euler's identity	phase constant (wave number) β	
attenuation factor	forcing function	phase lag and lead	
Biot-Savart law	fundamental dimensions	phase velocity (propagation velocity) u_p	
complex conjugate	instantaneous function	phasor	
complex number	law of conservation of electric charge	plane wave	
conductivity σ	LCD	principle of linear superposition	
constitutive parameters	liquid crystal	reference phase ϕ_0	
continuous periodic wave	lossless or lossy medium	relative permittivity or dielectric constant ϵ_r	
Coulomb's law	magnetic field intensity $\mathbf{H}$	SI system of units	
dielectric constant	magnetic flux density $\mathbf{B}$	static	
dynamic	magnetic force	transient wave	
electric dipole	magnetic permeability μ	velocity of light c	
electric field intensity $\mathbf{E}$	magnetostatics	wave amplitude	
electric flux density $\mathbf{D}$	microwave band	wave frequency f	
electric polarization	monochromatic	wave period T	
electrical force	nonmagnetic materials	waveform	
electrical permittivity ϵ	perfect conductor	wavelength λ	
electrodynamics	perfect dielectric		
electrostatics	periodic		

1.4 A wave traveling along a string is given by

$y(x,t) = 2\sin(4\pi t + 10\pi x)$ (cm),

where x is the distance along the string in meters and y is the vertical displacement. Determine: (a) the direction of wave travel, (b) the reference phase ϕ_0 , (c) the frequency, (d) the wavelength, and (e) the phase velocity.

1.5 Two waves, $y_1(t)$ and $y_2(t)$, have identical amplitudes and oscillate at the same frequency, but $y_2(t)$ leads $y_1(t)$ by a phase angle of 60° . If

$y_1(t) = 4\cos(2\pi \times 10^3 t)$,

write the expression appropriate for $y_2(t)$ and plot both functions over the time span from 0 to 2 ms.

*1.6 The height of an ocean wave is described by the function

$y(x,t) = 1.5\sin(0.5t - 0.6x)$ (m).

Determine the phase velocity and wavelength, and then sketch $y(x,t)$ at $t = 2$ s over the range from $x = 0$ to $x = 2\lambda$.

1.7 A wave traveling along a string in the $+x$ direction is given by

$y_1(x,t) = A\cos(\omega t - \beta x)$,

where $x = 0$ is the end of the string, which is tied rigidly to a wall, as shown in Fig. P1.7.

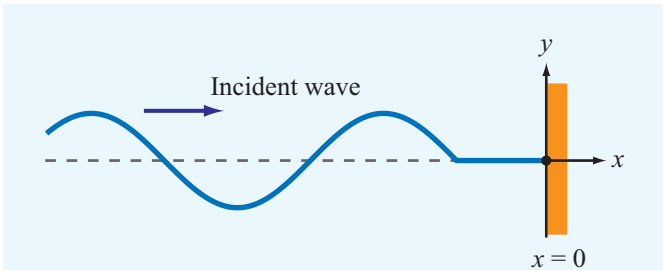


Figure P1.7 Wave on a string tied to a wall at $x = 0$ (Problem 1.7).

When wave $y_1(x,t)$ arrives at the wall, a reflected wave $y_2(x,t)$ is generated. Hence, at any location on the string, the vertical displacement y_s is the sum of the incident and reflected waves:

$y_s(x,t) = y_1(x,t) + y_2(x,t)$.

- (a) Write an expression for $y_2(x,t)$, keeping in mind its direction of travel and the fact that the end of the string cannot move.
- (b) Generate plots of $y_1(x,t)$, $y_2(x,t)$, and $y_s(x,t)$ versus x over the range $-2\lambda \leq x \leq 0$ at $\omega t = \pi/4$ and at $\omega t = \pi/2$.

重要术语 请给出下列术语的定义或解释其含义：

角频率 ω	电磁频谱	相位
衰减常数 α	欧拉恒等式	相位常数 (波数) β
衰减因子	激励函数	相位滞后与超前
毕奥-萨伐尔定律	基本量纲	相速度 (传播速度) u_p
复共轭	瞬时函数	相量
复数	电荷守恒定律	平面波
电导率 σ	液晶显示器 (LCD)	线性叠加原理
本构参数	液晶	参考相位 ϕ_0
连续周期波	无损介质或有损介质	相对介电率 ϵ_r (介电常数)
库仑定律	磁场强度 H	国际单位制 (SI)
介电常数	磁通密度 B	静态的
时变的	磁力	瞬态波
电偶极子	磁导率 μ	光速 c
电场强度 E	静磁学	波的振幅
电通量密度 D	微波波段	波的频率 f
电极化	单色的	波的周期 T
电场力	非磁性材料	波形
介电率 ϵ	理想导体	波长 λ
电动力学	理想电介质	
静电学	周期的	

1.4 沿弦传播的波为

$y(x,t) = 2\sin(4\pi t + 10\pi x)$ (cm),

其中, x 为沿弦的距离 (单位为 m), y 为竖直位移。求: (a) 波的传播方向; (b) 参考相位 ϕ_0 ; (c) 频率; (d) 波长; (e) 相速度。

1.5 两个波 $y_1(t)$ 与 $y_2(t)$ 的振幅、频率相同, $y_2(t)$ 的相位比 $y_1(t)$ 超前 60° 。若

$y_1(t) = 4\cos(2\pi \times 10^3 t)$,

写出 $y_2(t)$ 的表达式, 并绘制 0 至 2 ms 内这两个函数的曲线。

*1.6 海浪的高度由下式描述:

$y(x,t) = 1.5\sin(0.5t - 0.6x)$ (m).

求相速度及波长, 并画出 $t = 2$ s 时在 $0 \leq x \leq 2\lambda$ 范围内的波形 $y(x,t)$ 。

1.7 沿弦向 $+x$ 方向传播的波为

$y_1(x,t) = A\cos(\omega t - \beta x)$,

其中, $x = 0$ 为弦的固定端, 该端牢固地固定在墙上, 如图 P1.7 所示。

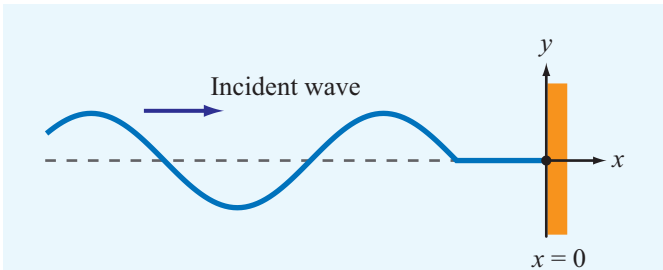


图 P1.7 弦在 $x = 0$ 处固定于墙上时的波 (习题 1.7)。图中箭头表示入射波。

当 $y_1(x,t)$ 到达墙壁时, 将产生反射波 $y_2(x,t)$ 。因此, 弦上任意位置的竖直位移 y_s 为入射波与反射波的位移之和:

$y_s(x,t) = y_1(x,t) + y_2(x,t)$.

(a) 写出 $y_2(x,t)$ 的表达式, 注意其传播方向以及弦的固定端不能移动这一条件。

(b) 分别在 $\omega t = \pi/4$ 和 $\omega t = \pi/2$ 时, 绘制 $-2\lambda \leq x \leq 0$ 范围内 $y_1(x,t)$ 、 $y_2(x,t)$ 与 $y_s(x,t)$ 随 x 变化的曲线。

校注: 术语表中的 angular velocity 在本章波动语境下指角频率 ω , 按该物理含义译出。

1.8 Two waves on a string are given by the functions:

$$y_1(x,t) = 4\cos(20t - 30x) \quad (\text{cm})$$

$$y_2(x,t) = -4\cos(20t + 30x) \quad (\text{cm}),$$

where x is in centimeters. The waves are said to interfere constructively when their superposition $|y_s| = |y_1 + y_2|$ is a maximum, and they interfere destructively when $|y_s|$ is a minimum.

- ***(a)** What are the directions of propagation of waves $y_1(x,t)$ and $y_2(x,t)$?
- (b)** At $t = (\pi/50)$ s, what location x do the two waves interfere constructively, and what is the corresponding value of $|y_s|$?
- (c)** At $t = (\pi/50)$ s, what location x do the two waves interfere destructively, and what is the corresponding value of $|y_s|$?

1.9 Find expressions for $y(x,t)$ for a sinusoidal wave traveling along a string in the negative x direction, given that $y_{\max} = 40$ cm, $\lambda = 30$ cm, $f = 10$ Hz, and

- (a)** $y(x,0) = 0$ at $x = 0$
- (b)** $y(x,0) = 0$ at $x = 3.75$ cm

***1.10** An oscillator that generates a sinusoidal wave on a string completes 20 vibrations in 50 s. The wave peak is observed to travel a distance of 2.8 m along the string in 5 s. What is the wavelength?

1.11 The vertical displacement of a string is given by the harmonic function:

$$y(x,t) = 2\cos(16\pi t - 20\pi x) \quad (\text{m}),$$

where x is the horizontal distance along the string in meters. Suppose a tiny particle were attached to the string at $x = 5$ cm. Obtain an expression for the vertical velocity of the particle as a function of time.

***1.12** Given two waves characterized by

$$y_1(t) = 3\cos\omega t,$$

$$y_2(t) = 3\sin(\omega t + 60^\circ),$$

does $y_2(t)$ lead or lag $y_1(t)$ and by what phase angle?

1.13 The voltage of an electromagnetic wave traveling on a transmission line is given by

$$v(z,t) = 5e^{-\alpha z}\sin(4\pi \times 10^9 t - 20\pi z) \quad (\text{V}),$$

where z is the distance in meters from the generator.

- (a)** Find the frequency, wavelength, and phase velocity of the wave.
- (b)** At $z = 2$ m, the amplitude of the wave was measured to be 2 V. Find α .

***1.14** A certain electromagnetic wave traveling in seawater was observed to have an amplitude of 98.02 (V/m) at a depth of 10 m, and an amplitude of 81.87 (V/m) at a depth of 100 m. What is the attenuation constant of seawater?

1.15 A laser beam traveling through fog was observed to have an intensity of $1\text{ }(\mu\text{W}/\text{m}^2)$ at a distance of 2 m from the laser gun and an intensity of $0.2\text{ }(\mu\text{W}/\text{m}^2)$ at a distance of 3 m. Given that the intensity of an electromagnetic wave is proportional to the square of its electric-field amplitude, find the attenuation constant α of fog.

1.16 In Module 1.2, set $f = 1$ Hz, $\lambda = 2$ cm, and $\alpha = 0.5$ Np/cm. Estimate the amplitude $y(x)$ of the blue wave at its first maximum (at $x = 0$) and at its second maximum (at $x = 2$ cm). Compute the ratio $y(2\text{ cm})/y(0)$. Does the result match your expectations given the values of f , λ , and α ?

Section 1-5: Complex Numbers

1.17 Complex numbers z_1 and z_2 are given as

$$z_1 = 3 - j2,$$

$$z_2 = -4 + j3.$$

- (a)** Express z_1 and z_2 in polar form.
- (b)** Find $|z_1|$ by first applying Eq. (1.41) and then by applying Eq. (1.43).
- ***(c)** Determine the product $z_1 z_2$ in polar form.
- (d)** Determine the ratio z_1/z_2 in polar form.
- (e)** Determine z_1^3 in polar form.

1.18 Evaluate each of the following complex numbers and express the result in rectangular form:

- (a)** $z_1 = 8e^{j\pi/3}$
- ***(b)** $z_2 = \sqrt{3}\text{ }e^{j3\pi/4}$
- (c)** $z_3 = 2e^{-j\pi/2}$
- (d)** $z_4 = j^3$
- (e)** $z_5 = j^{-4}$
- (f)** $z_6 = (1 - j)^3$
- (g)** $z_7 = (1 - j)^{1/2}$

1.8 弦上的两个波由以下函数给出:

$$y_1(x,t) = 4\cos(20t - 30x) \quad (\text{cm})$$

$$y_2(x,t) = -4\cos(20t + 30x) \quad (\text{cm}),$$

其中 x 的单位是厘米。当波的叠加 $|y_s| = |y_1 + y_2|$ 为最大值时, 波被称为相长干涉; 当 $|y_s|$ 为最小值时, 波被相消干涉。

***(a)** 波 $y_1(x,t)$ 和 $y_2(x,t)$ 的传播方向是什么? **(b)** 在 $t = (\pi/50)$ s 时, 两个波在什么位置 x 发生相长干涉, $|y_s|$ 对应的值是多少? **(c)** 在 $t = (\pi/50)$ s 处, 两个波在 x 的什么位置发生相消干涉, 对应的 $|y_s|$ 值是多少?

1.9 求出正弦波在负 x 方向上沿弦传播的 $y(x,t)$ 表达式, 假设 $y_{\max} = 40$ cm, $\lambda = 30$ cm, $f = 10$ Hz, 并且

- (a)** $y(x,0) = 0$ 在 $x = 0$
- (b)** $y(x,0) = 0$ 在 $x = 3.75$ cm *

1.10 在弦上产生正弦波的振荡器在 50 秒内完成 20 次振动。观察到波峰在 5 s 内沿着弦传播了 2.8 m 的距离。波长是多少?

1.11 弦的垂直位移由调和函数给出:

$$y(x,t) = 2\cos(16\pi t - 20\pi x) \quad (\text{m}),$$

其中 x 是沿绳索的水平距离 (以米为单位) 。假设一个微小的颗粒附着在 $x = 5$ 厘米处的绳子上。获得粒子垂直速度随时间变化的表达式。

*1.12 给定两个波, 其特征为

$$y_1(t) = 3\cos\omega t,$$

$$y_2(t) = 3\sin(\omega t + 60^\circ),$$

$y_2(t)$ 超前还是滞后 $y_1(t)$ 以及相位角是多少?

1.13 在传输线上传播的电磁波的电压由下式给出

$$v(z,t) = 5e^{-\alpha z}\sin(4\pi \times 10^9 t - 20\pi z) \quad (\text{V}),$$

其中 z 是距发电机的距离 (以米为单位) 。

(a) 求出波的频率、波长和相速度。 **(b)** 在 $z = 2$ m 处, 测得波的振幅为 2 V。求出 α 。

*1.14 某电磁波在海水中传播, 在10m深度处观测到振幅为 98.02 (V/m) , 在100m深度处振幅为81.87 (V/m) 。海水的衰减常数是多少?

1.15 穿过雾气的激光束在距离激光枪 2 m 处观察到强度为 $1\text{ }(\mu\text{W}/\text{m}^2)$, 在距离激光枪 3 m 处强度为 $0.2\text{ }(\mu\text{W}/\text{m}^2)$ 。假定电磁波的强度与其电场幅度的平方成正比, 求雾的衰减常数 α 。

1.16 在模块 1.2 中, 设置 $f = 1$ Hz、 $\lambda = 2$ cm 和 $\alpha = 0.5$ Np/cm。估计蓝色波在第一个最大值 ($x = 0$ 处) 和第二个最大值 ($x = 2$ cm 处) 的振幅 $y(x)$ 。计算比率 $y(2\text{ cm})/y(0)$ 。给定 f 、 λ 和 α 的值, 结果是否符合您的预期?

第 1-5 节: 复数

1.17 复数 z_1 和 z_2 给出为

$$z_1 = 3 - j2,$$

$$z_2 = -4 + j3.$$

- (a)** 以极坐标形式表示 z_1 和 z_2 。
- (b)** 首先应用方程求出 $|z_1|$ 。 (1.41) , 然后应用方程。(1.43) 。
- ***(c)** 确定极坐标形式的乘积 $z_1 z_2$ 。
- (d)** 确定极坐标形式的比率 z_1/z_2 。
- (e)** 确定极坐标形式的 z_1^3 。

1.18 计算下列每个复数并用直角形式表示结果:

- (a)** $z_1 = 8e^{j\pi/3}$
- ***(b)** $z_2 = \sqrt{3}\text{ }e^{j3\pi/4}$
- (c)** $z_3 = 2e^{-j\pi/2}$
- (d)** $z_4 = j^3$
- (e)** $z_5 = j^{-4}$
- (f)** $z_6 = (1 - j)^3$
- (g)** $z_7 = (1 - j)^{1/2}$

1.19 Complex numbers z_1 and z_2 are given by

$$z_1 = -3 + j2,$$

$$z_2 = 1 - j2.$$

Determine (a) $z_1 z_2$, (b) z_1/z_2^* , (c) z_1^2 , and (d) $z_1 z_1^*$ all in polar form.

1.20 If $z = -2 + j4$, determine the following quantities in polar form:

- (a) $1/z$
 (b) z^3
 * (c) $|z|^2$
 (d) $\Im\{z\}$
 (e) $\Im\{z^*\}$

1.21 Find complex numbers $t = z_1 + z_2$ and $s = z_1 - z_2$, both in polar form, for each of the following pairs:

- (a) $z_1 = 2 + j3$ and $z_2 = 1 - j2$
 (b) $z_1 = 3$ and $z_2 = -j3$
 (c) $z_1 = 3/\underline{30^\circ}$ and $z_2 = 3/\underline{-30^\circ}$
 * (d) $z_1 = 3/\underline{30^\circ}$ and $z_2 = 3/\underline{-150^\circ}$

1.22 Complex numbers z_1 and z_2 are given by

$$z_1 = 5/\underline{-60^\circ},$$

$$z_2 = 4/\underline{45^\circ}.$$

- (a) Determine the product $z_1 z_2$ in polar form.
 (b) Determine the product $z_1 z_2^*$ in polar form.
 (c) Determine the ratio z_1/z_2 in polar form.
 (d) Determine the ratio z_1^*/z_2^* in polar form.
 (e) Determine $\sqrt{z_1}$ in polar form.

* **1.23** If $z = 3 - j5$, find the value of $\ln(z)$.

1.24 If $z = 3e^{j\pi/6}$, find the value of e^z .

1.25 If $z = 3 - j4$, find the value of e^z .

Section 1-6: Phasors

* **1.26** A voltage source given by

$$v_s(t) = 25 \cos(2\pi \times 10^3 t - 30^\circ) \quad (\text{V})$$

is connected to a series RC load as shown in **Fig. 1-20**. If $R = 1 \text{ M}\Omega$ and $C = 200 \text{ pF}$, obtain an expression for the voltage across the capacitor, $v_c(t)$.

1.27 Find the instantaneous time sinusoidal functions corresponding to the following phasors:

- (a) $\tilde{V} = -5e^{j\pi/3} \text{ (V)}$
 (b) $\tilde{V} = j6e^{-j\pi/4} \text{ (V)}$
 (c) $\tilde{I} = (6 + j8) \text{ (A)}$
 * (d) $\tilde{I} = -3 + j2 \text{ (A)}$
 (e) $\tilde{I} = j \text{ (A)}$
 (f) $\tilde{I} = 2e^{j\pi/6} \text{ (A)}$

1.28 Find the phasors of the following time functions:

- (a) $v(t) = 9 \cos(\omega t - \pi/3) \text{ (V)}$
 (b) $v(t) = 12 \sin(\omega t + \pi/4) \text{ (V)}$
 (c) $i(x, t) = 5e^{-3x} \sin(\omega t + \pi/6) \text{ (A)}$
 * (d) $i(t) = -2 \cos(\omega t + 3\pi/4) \text{ (A)}$
 (e) $i(t) = 4 \sin(\omega t + \pi/3) + 3 \cos(\omega t - \pi/6) \text{ (A)}$

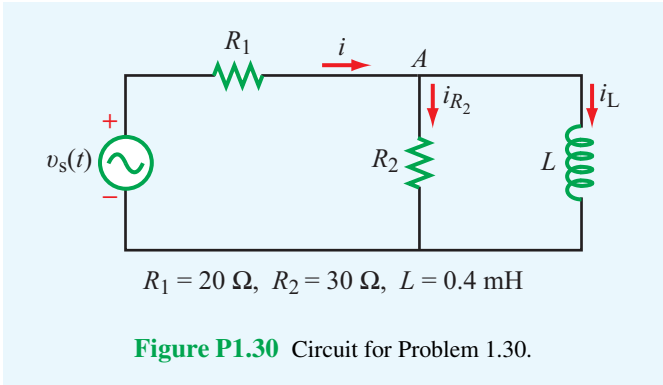
1.29 A series RLC circuit is connected to a generator with a voltage $v_s(t) = V_0 \cos(\omega t + \pi/3) \text{ (V)}$.

- (a) Write the voltage loop equation in terms of the current $i(t)$, R , L , C , and $v_s(t)$.
 (b) Obtain the corresponding phasor-domain equation.
 (c) Solve the equation to obtain an expression for the phasor current $\tilde{I}$.

1.30 The voltage source of the circuit shown in **Fig. P1.30** is given by

$$v_s(t) = 25 \cos(4 \times 10^4 t - 45^\circ) \quad (\text{V}).$$

Obtain an expression for the current flowing through the inductor, $i_L(t)$.



1.31 Repeat Problem 1.30, but this time determine current $i_{R_2}(t)$.

习题 45

1.19 复数 z_1 和 z_2 为

$$z_1 = -3 + j2,$$

$$z_2 = 1 - j2.$$

用极坐标形式表示: (a) $z_1 z_2$; (b) z_1/z_2^* ; (c) z_1^2 ; (d) $z_1 z_1^*$. 星号表示复共轭。

1.20 若 $z = -2 + j4$, 求下列各量的极坐标形式:

- (a) $1/z$
 (b) z^3
 * (c) $|z|^2$
 (d) $\Im\{z\}$
 (e) $\Im\{z^*\}$

1.21 对下列各组复数, 求 $t = z_1 + z_2$ 和 $s = z_1 - z_2$, 并用极坐标形式表示:

- (a) $z_1 = 2 + j3$, $z_2 = 1 - j2$
 (b) $z_1 = 3$, $z_2 = -j3$
 (c) $z_1 = 3\angle 30^\circ$, $z_2 = 3\angle -30^\circ$
 * (d) $z_1 = 3\angle 30^\circ$, $z_2 = 3\angle -150^\circ$

1.22 复数 z_1 和 z_2 为

$$z_1 = 5/\underline{-60^\circ},$$

$$z_2 = 4/\underline{45^\circ}.$$

- (a) 用极坐标形式表示乘积 $z_1 z_2$.
 (b) 用极坐标形式表示乘积 $z_1 z_2^*$.
 (c) 用极坐标形式表示比值 z_1/z_2 .
 (d) 用极坐标形式表示比值 z_1^*/z_2^* .
 (e) 用极坐标形式表示 $\sqrt{z_1}$.

* 1.23 若 $z = 3 - j5$, 求 $\ln(z)$.

1.24 若 $z = 3e^{j\pi/6}$, 求 e^z .

1.25 若 $z = 3 - j4$, 求 e^z .

相量 (第 1-7 节)

* 1.26 电压源

$$v_s(t) = 25 \cos(2\pi \times 10^3 t - 30^\circ) \quad (\text{V})$$

接至图 1-20 所示的串联 RC 负载。若 $R = 1 \text{ M}\Omega$ 、 $C = 200 \text{ pF}$, 求电容器两端电压 $v_c(t)$ 的表达式。

1.27 求与下列相量对应的瞬时正弦时间函数:

- (a) $\tilde{V} = -5e^{j\pi/3} \text{ (V)}$
 (b) $\tilde{V} = j6e^{-j\pi/4} \text{ (V)}$
 (c) $\tilde{I} = (6 + j8) \text{ (A)}$
 * (d) $\tilde{I} = -3 + j2 \text{ (A)}$
 (e) $\tilde{I} = j \text{ (A)}$
 (f) $\tilde{I} = 2e^{j\pi/6} \text{ (A)}$

1.28 求下列时间函数对应的相量:

- (a) $v(t) = 9 \cos(\omega t - \pi/3) \text{ (V)}$
 (b) $v(t) = 12 \sin(\omega t + \pi/4) \text{ (V)}$
 (c) $i(x, t) = 5e^{-3x} \sin(\omega t + \pi/6) \text{ (A)}$
 * (d) $i(t) = -2 \cos(\omega t + 3\pi/4) \text{ (A)}$
 (e) $i(t) = 4 \sin(\omega t + \pi/3) + 3 \cos(\omega t - \pi/6) \text{ (A)}$

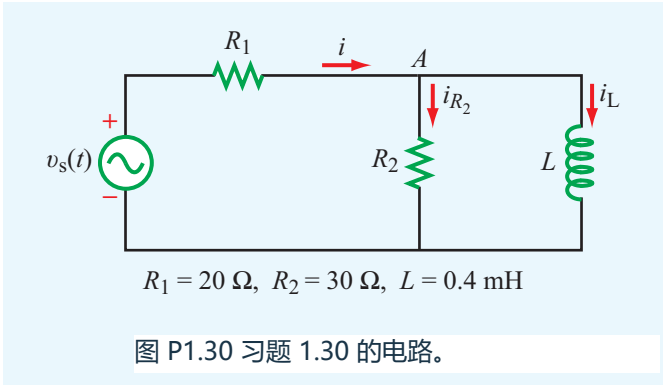
1.29 串联 RLC 电路接至电压为 $v_s(t) = V_0 \cos(\omega t + \pi/3) \text{ (V)}$ 的电源。

- (a) 用电流 $i(t)$ 、 R 、 L 、 C 和 $v_s(t)$ 写出回路电压方程。
 (b) 写出对应的相量域方程。
 (c) 解此方程, 求出电流相量。

1.30 图 P1.30 所示电路的电压源为

$$v_s(t) = 25 \cos(4 \times 10^4 t - 45^\circ) \quad (\text{V}).$$

求流过电感器的电流 $i_L(t)$ 的表达式。



1.31 对习题 1.30 的电路, 求电流 $i_{R_2}(t)$ 。